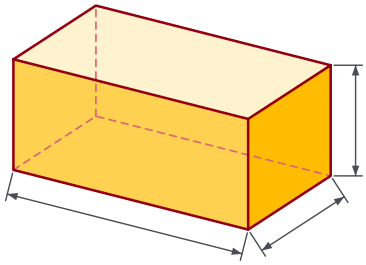
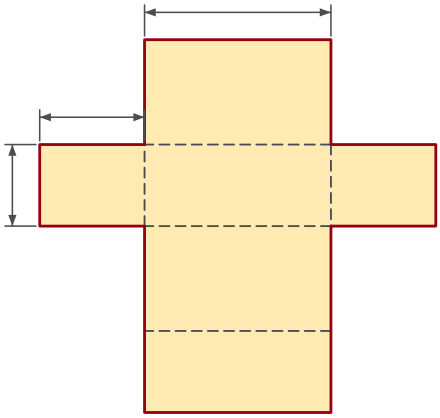
ACCESS TRADES

# Trades Numeracy Exercise Book

Practice for every chapter of the Trades Numeracy Textbook: skills, shop problems and Shape Canvas drawings, in metric and imperial. For fabrication, pipe trades, millwright and carpentry pre-apprentices.





ACCESS

Skills for Success: Numeracy
Answers are in the Answer Key

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# How to use this book

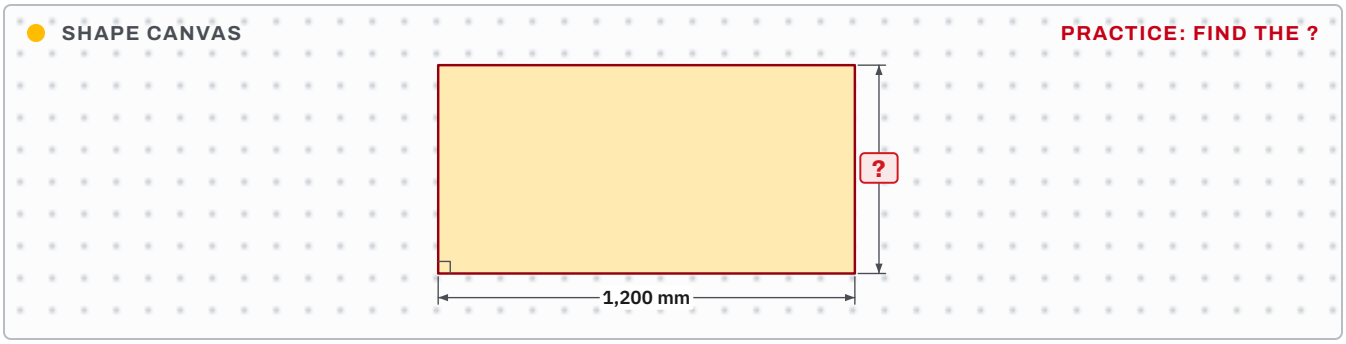
Each chapter of this book matches the chapter with the same number in the Trades Numeracy Textbook. Work through the textbook chapter first, then use these exercises for more practice.

Every chapter starts with three sets. Set A practises the skills one at a time. Set B puts them to work in shop problems. Set C uses Shape Canvas drawings: a red ? marks a hidden size to find, dashed lines break a part into simple shapes, and a solid shown beside its flat pattern is unfolded into its net.

Show your work in the working box under each problem and write the answer on the answer line. Estimate first, then use your calculator. The keys for the TI-30Xa and the Sharp EL-531XT are shown in the textbook. Answers to every problem are in the Answer Key.

Some chapters have more sets after Set C. A set marked NO CALCULATOR is done by hand, the way the ACCESS math assessment is written. A set titled Carpentry and the BCBC puts the chapter to work in carpentry, the fourth trade in these books alongside fabrication, pipe trades and millwright. Its code values are quoted from the BC Building Code 2024 with the Article, Sentence and page, as the textbook explains on its page Using the BC Building Code.

After Chapter 13 come two appendices. Appendix A, Kitchen Math, practises recipe scaling, cooking measures and food yield percent. Appendix B is a Practice Assessment: 40 problems by hand, in the same skill order as the ACCESS math assessment.



# 1 Whole Numbers

EXERCISES

Practice with place value, rounding and estimating, the four operations, averages and multi-step shop problems. Estimate each answer first, then check it with your calculator.

**EXERCISE 1A**

## Skills

Place value, rounding, estimating and the four operations

Name _____

Date _____

**1** Write 4,072 in words.

WORKING

Answer _____

**2** Write in digits: fifteen thousand three hundred eight.

WORKING

Answer _____

**3** What is the 7 worth in 37,215?

WORKING

Answer _____

**4** Put in order, smallest first: 3,409, 3,490, 3,049, 3,940.

WORKING

Answer _____

**5** Round 3,462 to the nearest ten.

WORKING

Answer _____

**6** Round 8,951 to the nearest hundred.

WORKING

Answer _____

**7** Round 14,500 to the nearest thousand.

WORKING

Answer _____

**8** Round 795 to the nearest ten.

WORKING

Answer _____

**9** Estimate 1,180 + 2,940 + 615 by rounding each number to the nearest hundred.

WORKING

Answer _____

**10** Estimate 48 × 21 by rounding each number to the nearest ten.

WORKING

Answer _____

**11** 3,275 + 1,948

WORKING

Answer _____

**12** 509 + 2,716 + 88

WORKING

Answer _____

**13** 7,000 – 2,368

WORKING

Answer _____

**14** 10,250 – 4,875

WORKING

Answer _____

**15** 32 × 15

WORKING

Answer _____

**16** 215 × 24

WORKING

Answer _____

**17** 2,688 ÷ 12

WORKING

Answer _____

**18** Divide 5,000 by 320. Give the whole-number quotient and the remainder.

WORKING

Answer _____

**19** Find the average of 64, 58, 71 and 67.

WORKING

Answer _____

**20** Check 7,000 – 2,368 by adding. What should you get?

WORKING

Answer _____

EXERCISE 1B

## Shop problems

Whole numbers on the job, in metric and imperial

Name _____

Date _____

**1** Three pieces, 1,350 mm, 980 mm, 2,145 mm, are cut from a 6,000 mm length of flat bar. Ignore the saw cuts. How long is the offcut?

WORKING

Answer _____

**2** A plumber measures five pipe runs: 12 ft, 9 ft, 15 ft, 7 ft, 18 ft. The supplier sells this pipe in 21 ft lengths. Ignore fittings and waste. How many feet of pipe are needed, and how many 21 ft lengths must be bought?

WORKING

Answer _____

**3** A millwright sets 14 baseplates. Each baseplate takes 8 anchor bolts. Anchor bolts come in boxes of 25. How many bolts are needed, and how many boxes?

WORKING

Answer _____

**4** A pipe job has 9 flanged joints. Each joint takes 8 bolts and each bolt takes 2 nuts. How many nuts are needed?

WORKING

Answer _____

**5** Lengths of 18 in are cut from a 20 ft bar of angle (20 ft = 240 in). Ignore the saw cuts. How many pieces can be cut, and how long is the offcut?

WORKING

Answer _____

**6** A 48 in bar has 7 holes in a straight row. The end holes are 3 in from the ends and the holes are evenly spaced. Find the spacing, centre to centre.

WORKING

Answer _____

**7** A welder logs 7, 9, 8, 6, 10 hours over five days. What is the average number of hours per day?

WORKING

Answer _____

**8** A machine makes 45 parts an hour. It runs 3 shifts of 8 hours. How many parts does it make? Estimate first, using 50 parts an hour.

WORKING

Answer _____

**9** A crew uses 18 welding rods a day for 12 days. Rods come in boxes of 50. How many boxes do they need, and how many rods are left over?

WORKING

Answer _____

**EXERCISE 1C**

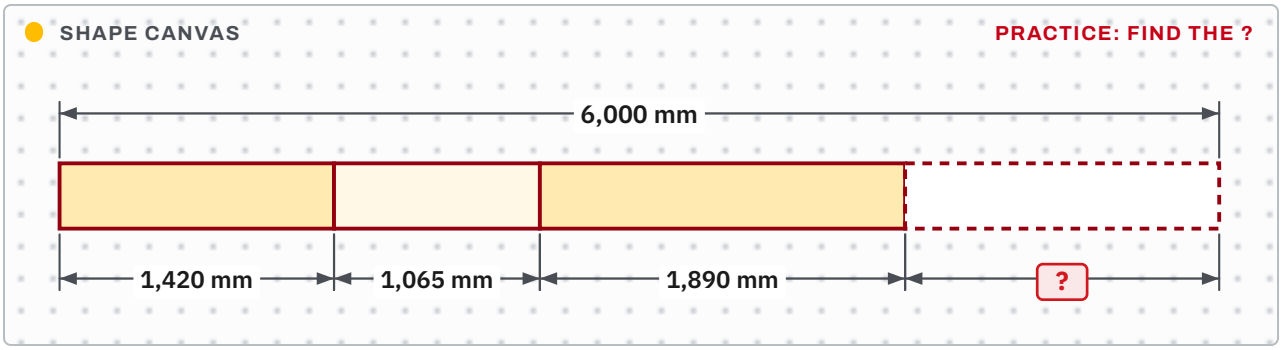
# Shape Canvas

Find each red ? on the drawing

Name _____

Date _____

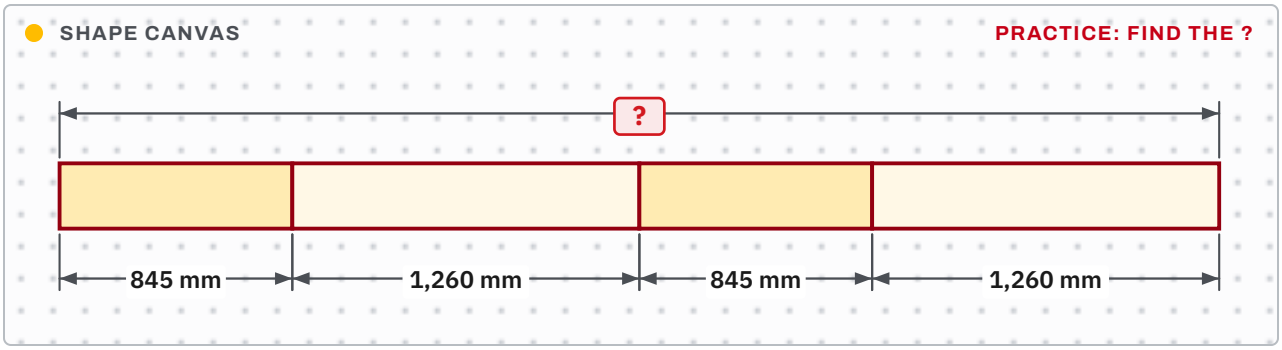
**1** Three pieces are cut from a 6,000 mm bar. Ignore the saw cuts. Find the offcut.




WORKING

Answer _____

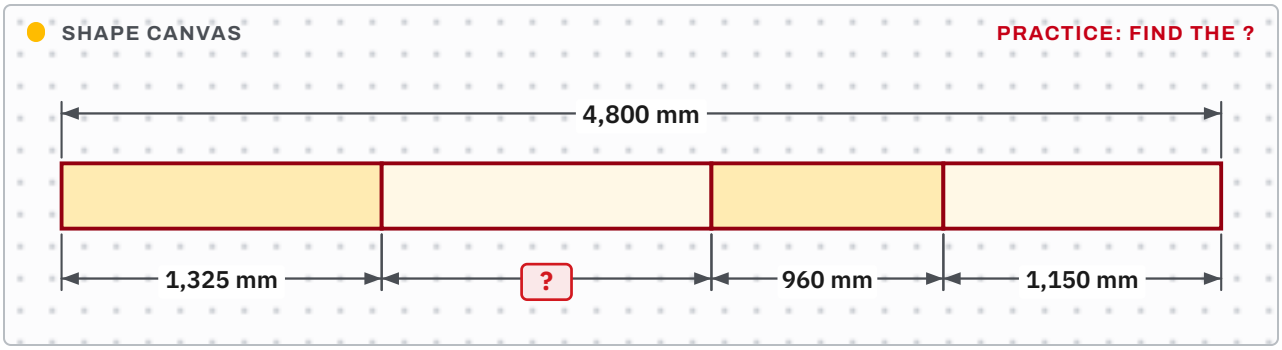
**2** Find the overall length.




WORKING

Answer _____

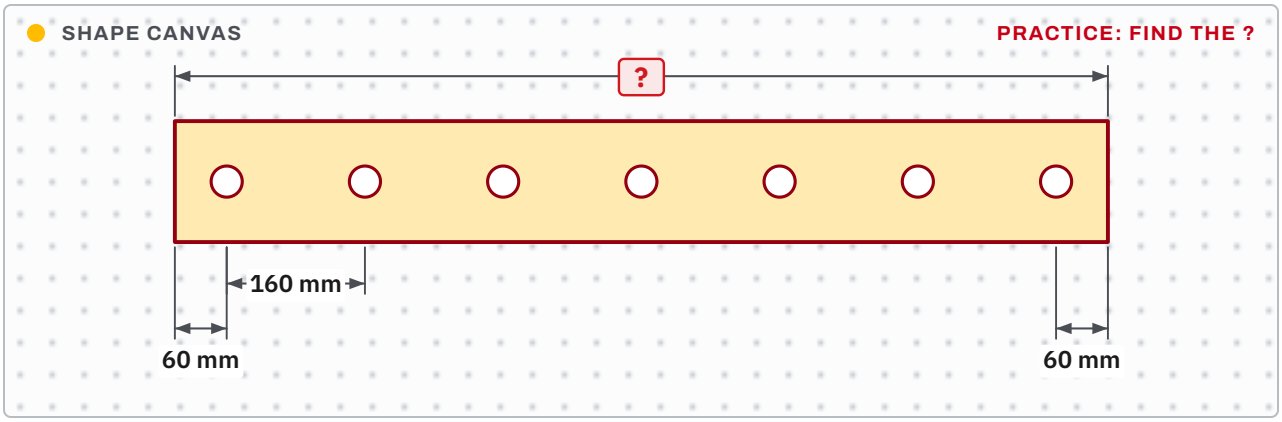
**3** The overall length is 4,800 mm. Find the hidden piece.




WORKING

Answer _____

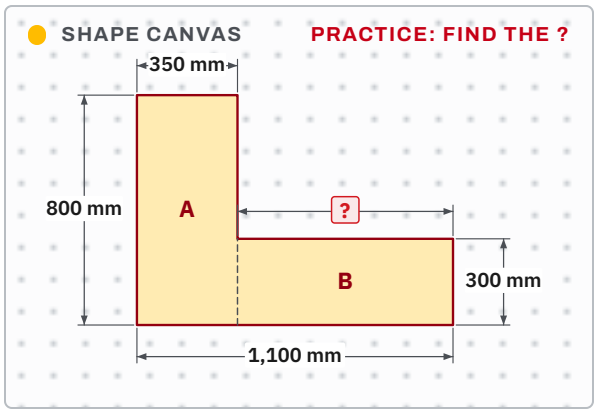
**4** Seven holes are evenly spaced 160 mm apart along a plate. The end holes are 60 mm from the ends. Find the length of the plate.




WORKING

Answer _____

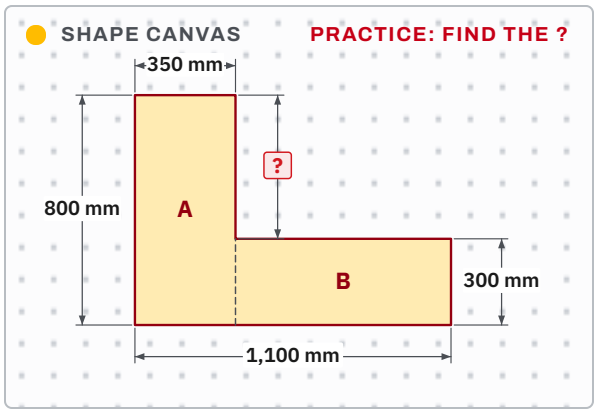
**5** Find the hidden width. The dashed line splits the plate into A and B.




WORKING

Answer _____

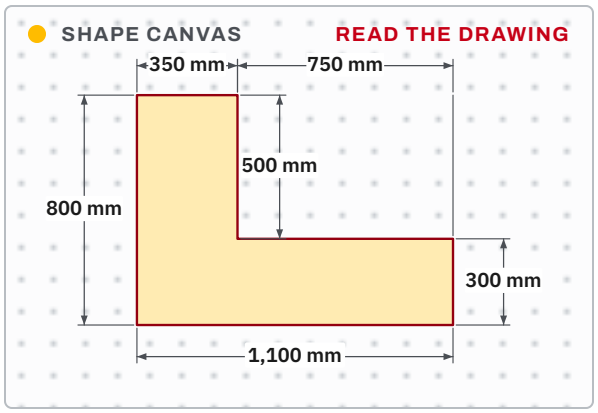
**6** How far does A rise above the top of B? Find the hidden height.




WORKING

Answer _____

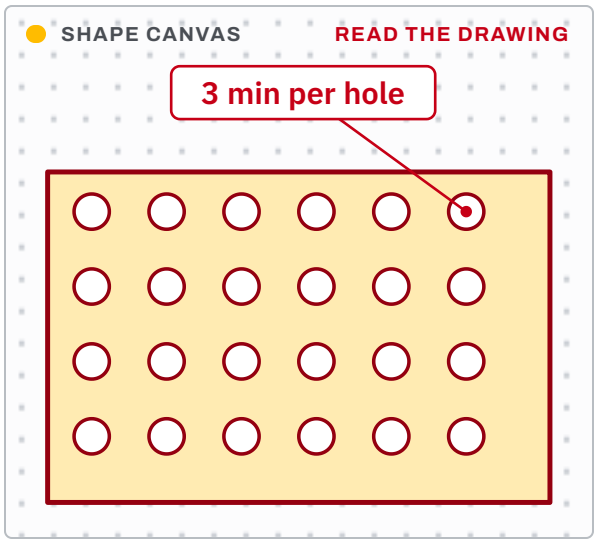
**7** Find the distance around the plate: add all six edges.




WORKING

Answer _____

**8** Count the holes with one multiplication. How long does it take to drill all the holes?




WORKING

Answer _____

**9** Change one value: each of the three pieces in problem 1 is cut 25 mm longer. How much shorter is the offcut now?

WORKING

Answer _____

**EXERCISE 1D**

# By hand

Column adding and subtracting, long multiplication and long division. No calculator.

Name _____

Date _____

Work each problem on paper in columns. Show the carries, the borrows and the partial products. Write each remainder with an r, as in 25 r 3.

**1** 4,786 + 3,958 + 607

WORKING

Answer _____

**2** 10,000 – 3,476

WORKING

Answer _____

**3** 5,003 – 1,867

WORKING

Answer _____

**4** 64 × 37

WORKING

Answer _____

**5** 506 × 23

WORKING

Answer _____

**6** 348 × 275

WORKING

Answer _____

**7** 709 × 406

WORKING

Answer _____

**8** 7,284 ÷ 6

WORKING

Answer _____

**9** 9,135 ÷ 8

WORKING

Answer _____

**10** 6,048 ÷ 24

WORKING

Answer _____

**11** 8,500 ÷ 32

WORKING

Answer _____

**12** 13,580 ÷ 45

WORKING

Answer _____

EXERCISE 1K

# Carpentry and the BCBC

Studs, joists, plates, sheets and time on a framing job

Name _____

Date _____

Every wall has a stud at each end, and spacings are on centre (o.c.). Round the number of spaces up when a spacing is a maximum. Code values are from BC Building Code 2024, Division B.

**1** A wall is 7,200 mm long with studs at 400 mm o.c. How many studs? How many fewer studs would it need at 600 mm o.c.?

WORKING

Answer _____

**2** Table 9.23.10.1 (Article 9.23.10.1, Sentence (1), page 9-182) lists exterior walls that support a roof with or without attic storage. For 38 × 89 studs, what is the maximum stud spacing and the maximum unsupported height?

WORKING

Answer _____

**3** Table 9.23.10.1 allows 38 × 89 studs at no more than 300 mm o.c. in an exterior wall that supports a roof plus 2 floors (Article 9.23.10.1, page 9-182). How many studs does a 4,500 mm length of that wall need?

WORKING

Answer _____

**4** Table 9.23.10.1 allows 38 × 64 studs in an exterior wall that supports a roof only at no more than 400 mm o.c., with a maximum unsupported height of 2.4 m. A wall 3,900 mm long is laid out at 400 mm from one end. How many studs, and how wide is the last space?

WORKING

Answer _____

**5** Table 9.29.5.3 (Article 9.29.5.3, Sentence (1), page 9-275) says 12.7 mm gypsum board on a ceiling with a painted finish, laid parallel to the framing, needs supports no more than 400 mm o.c. A ceiling is 5,000 mm long. How many ceiling joists, with one at each end?

WORKING

Answer _____

**6** You need 18 cripple studs, each 457 mm long, cut from boards 2,438 mm long. Allow 3 mm of kerf for each piece. How many pieces come from one board, and how many boards do you need?

WORKING

Answer _____

**7** A 7,200 mm wall has one bottom plate and a double top plate. Plate stock comes in 4,880 mm lengths. How many millimetres of plate are needed, and how many lengths must be bought? Ignore the joints.

WORKING

Answer _____

**8** The 7,200 mm wall is sheathed with sheets 1,220 mm wide, in 2 rows. Ignore openings. How many sheets?

WORKING

Answer _____

**9** A framing crew of 4 starts at 7:15 a.m. and finishes at 3:45 p.m., with a 30 minute unpaid lunch. How many paid hours does each worker have, and how many paid hours does the crew have in total?

WORKING

Answer _____

**10** Five walls need 13, 19, 13, 9, 16 studs. Find the mean, the median and the mode.

WORKING

Answer _____

# 2 Your Calculator: TI-30Xa and Sharp EL-531XT

EXERCISES

Practice with the keys of both calculators: writing key sequences, second functions, negative numbers, powers and roots, π, fractions, percent, memory and the trig keys. Start every problem with ON/C ON/C and DEG on the display.

**EXERCISE 2A**

## Skills

Write the keys for both calculators, then the answer

Name _____

Date _____

**1** 3,640 + 1,295 − 788

WORKING

Answer _____

**2** $23^2$

WORKING

Answer _____

**3** $\sqrt{1{,}369}$

WORKING

Answer _____

**4** $7^3$

WORKING

Answer _____

**5** The cube root of 343

WORKING

Answer _____

**6** (−15) × 6 + 40

WORKING

Answer _____

**7** $3^5$, using the $y^x$ key

WORKING

Answer _____

**8** π × 90, to two decimal places

WORKING

Answer _____

**9** $2\frac{3}{8} + 1\frac{7}{16}$, using the fraction key. Write the answer as the display shows it.

WORKING

Answer _____

**10** Change $\frac{7}{8}$ to a decimal.

WORKING

Answer _____

**11** 18% of 450

WORKING

Answer _____

**12** 1,250 plus 8%

WORKING

Answer _____

**13** sin 52°, to four decimal places

WORKING

Answer _____

**14** The angle whose tangent is 0.6, to two decimal places

WORKING

Answer _____

**15** Store 640 + 385 in memory, clear the display, then find 2,400 minus the stored number.

WORKING

Answer _____

**16** 1 ÷ 8, using the 1/x key ($x^{-1}$ on the Sharp)

WORKING

Answer _____

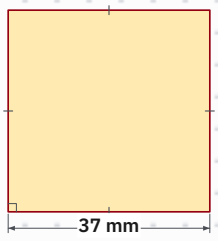
**17** A learner needs (−25) + 9 and presses the minus key to make the 25 negative. Which key should they use instead on each calculator? What is the answer?

WORKING

Answer _____

**18** Find the area of the square with the $x^2$ key.

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

EXERCISE 2B

# Shop problems

Use the calculator keys on the job

Name _____

Date _____

**1** A pipe has an outside diameter of 114.3 mm. The distance around it is π × D. Find it to one decimal place.

WORKING

Answer _____

**2** The round lid of a tank has a radius of 600 mm. Its area is π × r². Find it to the nearest square millimetre.

WORKING

Answer _____

**3** Three pieces, each $14\frac{5}{8}$ in long, are cut from a 72 in bar. Ignore the saw cuts. Use the fraction key to find the offcut: $72 - 3 \times 14\frac{5}{8}$.

WORKING

Answer _____

**4** A job needs 380 m of cable. The order adds 12% for waste. Use the % key to find the length to order.

WORKING

Answer _____

**5** Each frame has four welds: 1,240 mm, 860 mm, 1,240 mm and 860 mm. Store the weld length for one frame in memory, then use it to find the weld length for 18 frames.

WORKING

Answer _____

**6** The diagonal of a 1,200 mm by 900 mm frame is $\sqrt{(1{,}200^2 + 900^2)}$. Find it. Write the keys for both calculators.

WORKING

Answer _____

**7** A ramp rises 450 mm over a run of 2,400 mm. Its angle is the angle whose tangent is 450 ÷ 2,400. Find it to one decimal place. Check that DEG is on the display first.

WORKING

Answer _____

**8** A gauge reads −11. The setting it should read is 4. Find −11 − 4 to see how far off it is. Use the change-sign key.

WORKING

Answer _____

**EXERCISE 2C**

# Shape Canvas

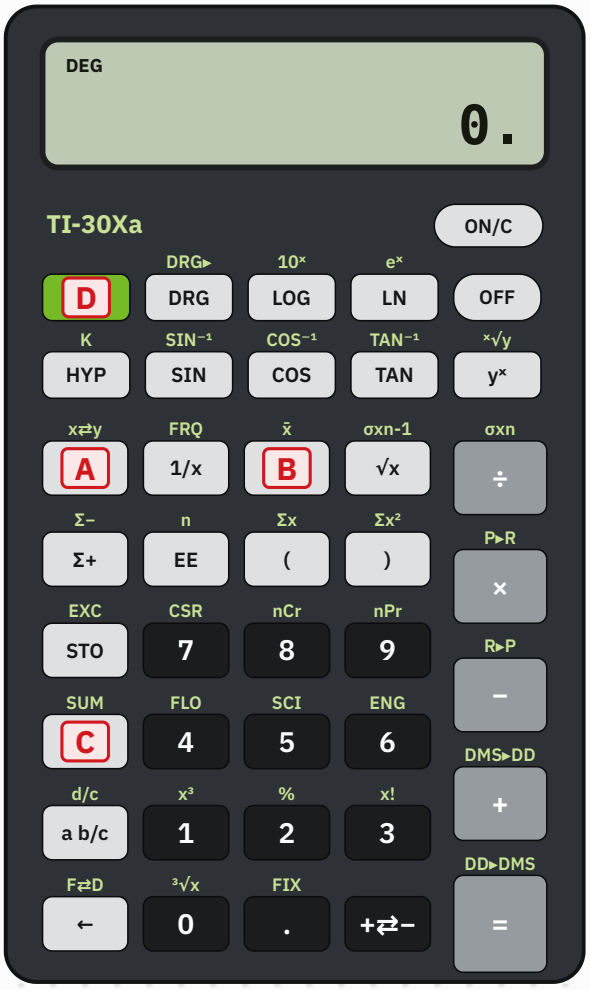
Keys, displays and drawings

Name _____

Date _____

**1** Name keys A to H and say what each one does. A to D are on the TI-30Xa, E to H on the Sharp EL-531XT.

SHAPE CANVAS — NAME THE KEYS





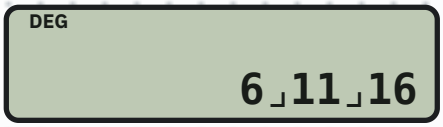


WORKING

Answer _____

**2** Write the number this display shows.

SHAPE CANVAS READ IT




TI-30Xa

WORKING Answer _____

**3** Write the number this display shows, with a comma.

SHAPE CANVAS READ IT




Sharp EL-531XT

WORKING Answer _____

**4** What calculation was keyed? Write it with an exponent.

SHAPE CANVAS READ IT




Sharp EL-531XT

WORKING Answer _____

**5** The small number at the right is an exponent of 10. Write the number in full, with commas.

SHAPE CANVAS READ IT




TI-30Xa

WORKING Answer _____

**6** This display shows GRAD. How many presses of DRG bring back DEG?

SHAPE CANVAS READ IT

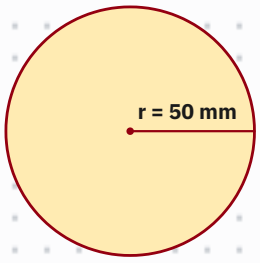



Sharp EL-531XT

WORKING Answer _____

**7** Find the distance around the circle, π × D, to two decimal places. Then change one value: if the diameter doubles, what is the new distance around?

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

EXERCISE 2K

# Carpentry and the BCBC

Tape lengths in feet, inches and fractions on both calculators

Name _____

Date _____

**1** Change $9' \; 4\frac{3}{4}''$ to inches.

WORKING

Answer _____

**2** Change $14' \; 11\frac{5}{8}''$ to inches.

WORKING

Answer _____

**3** Two lengths of baseboard, $5' \; 6\frac{3}{16}''$ and $3' \; 10\frac{15}{16}''$, are butted end to end. Find the total in feet and inches.

WORKING

Answer _____

**4** A piece $7' \; 9\frac{3}{8}''$ long is cut from a 16′ board. Ignore the saw cut. Find the piece left over in feet and inches. Put the length you take away in brackets.

WORKING

Answer _____

**5** A $10' \; 4\frac{1}{2}''$ board is cut into 3 equal shelves. Ignore the saw cuts. How long is each shelf?

WORKING

Answer _____

**6** Change $8' \; 1\frac{1}{8}''$ to millimetres, to the nearest millimetre. One inch is exactly 25.4 mm.

WORKING

Answer _____

**7** Two walls are $12' \; 6\frac{1}{2}''$ and $9' \; 11\frac{3}{4}''$ long. Each wall gets a top plate and a bottom plate. Find the total plate length in feet and inches.

WORKING

Answer _____

**8** A stair has 15 risers, each $7\frac{1}{4}$″ high. Find the total rise in feet and inches.

WORKING

Answer _____

**9** A stair has 14 treads, each with a unit run of $10\frac{1}{2}$″. Find the total run in feet and inches.

WORKING

Answer _____

**10** Table 9.8.4.2 of the BC Building Code (Sentence 9.8.4.2.(1), page 9-31) gives a minimum unit run of 255 mm for a private stair. Change 255 mm to inches, to the nearest $\frac{1}{16}$ in. The metric value stays the code limit.

WORKING

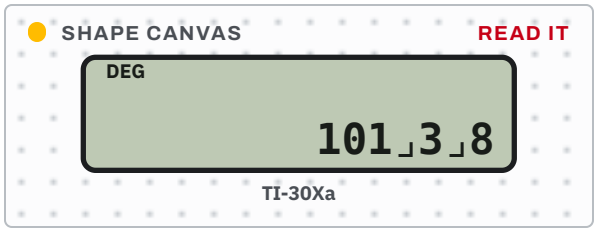
Answer _____

**11** Article 9.8.3.3, Sentence (1) (page 9-30), limits the height of one flight of stairs to 3.7 m, which is 3,700 mm. Change 3,700 mm to feet and inches. Round down to the $\frac{1}{16}$ in below, so the inch value is not over the limit.

WORKING

Answer _____

**12** A tape length was changed to inches on the TI-30Xa. Write the length in feet and inches.




WORKING

Answer _____

**13** A learner writes 8′ 6″ as 8.6 ft. What is wrong? Find the length in inches and in decimal feet.

WORKING

Answer _____

# 3 Order of Operations

EXERCISES

Practice with BEDMAS: brackets, exponents and roots, × and ÷ left to right, + and − left to right, and the fraction bar. Work each problem on paper first, then key it so the calculator follows the same order.

**EXERCISE 3A**

## Skills

Work each one out with BEDMAS

Name _____

Date _____

**1** $9 + 6 \times 4$

WORKING

Answer _____

**2** $50 - 18 \div 3$

WORKING

Answer _____

**3** $7 \times 8 - 6 \times 5$

WORKING

Answer _____

**4** $(15 - 7) \times 4$

WORKING

Answer _____

**5** $36 \div (4 + 5)$

WORKING

Answer _____

**6** $[30 - (8 + 7)] \div 5$

WORKING

Answer _____

**7** $2 \times [12 + (9 - 3)]$

WORKING

Answer _____

**8** $6^2 - 4 \times 5$

WORKING

Answer _____

**9** $(6 - 4)^2 \times 5$

WORKING

Answer _____

**10** $3 \times 4^2$

WORKING

Answer _____

**11** $\sqrt{(144 - 44)}$

WORKING

Answer _____

**12** $\sqrt{49} + \sqrt{81}$

WORKING

Answer _____

**13** $72 \div 6 \div 3$

WORKING

Answer _____

**14** $15 \div 3 \times 5$

WORKING

Answer _____

**15** $40 - 15 + 5$

WORKING

Answer _____

**16** $2 + 3^3$

WORKING

Answer _____

**17** $\frac{56 + 28}{9 - 2}$

WORKING

Answer _____

**18** $\frac{3 \times 15 + 5}{10 \times 2}$

WORKING

Answer _____

**19** Add one pair of brackets to make this true: $4 + 6 \times 3 = 30$.

WORKING

Answer _____

**20** Add one pair of brackets to make this true: $24 \div 6 - 2 = 6$.

WORKING

Answer _____

EXERCISE 3B

# Shop problems

Write one calculation for each job, then work it out

Name _____

Date _____

**1** A spool piece has 6 flanges with 8 bolts each. You also want 4 spare bolts. Write one calculation for the bolts needed and work it out.

WORKING

Answer _____

**2** A rectangular guard frame is 1,350 mm by 875 mm. The distance around it is $2 \times (1{,}350 + 875)$. Find it.

WORKING

Answer _____

**3** Four 27 in pieces and two 15 in pieces are cut from a 144 in bar. Ignore the saw cuts. The offcut is $144 - (4 \times 27 + 2 \times 15)$. Find it.

WORKING

Answer _____

**4** Three pipe spools measure 3,250 mm, 2,980 mm and 3,105 mm. Their average length is $\frac{3{,}250 + 2{,}980 + 3{,}105}{3}$. Find it to two decimal places.

WORKING

Answer _____

**5** A 60 in bar has 5 evenly spaced holes, with the end holes 4 in from the ends. The spacing is $\frac{60 - 2 \times 4}{4}$. Find it.

WORKING

Answer _____

**6** Three pipe pieces of 1,200 mm, 850 mm and 640 mm go into a run with 6 fittings. For this job each fitting takes up 25 mm of the run. The length of the run is (1,200 + 850 + 640) + 6 × 25. Find it.

WORKING

Answer _____

**7** A crew of 3 works 8 hours and a crew of 2 works 10 hours. The total hours are 3 × 8 + 2 × 10. A learner keys (3 × 8 + 2) × 10 instead. Find both answers and say which is right.

WORKING

Answer _____

**8** A learner finds a hole spacing by keying 1,500 − 300 ÷ 4 = and gets 1,425. The spacing should be $\frac{1{,}500 - 300}{4}$. What went wrong? What are the right keys and the right answer?

WORKING

Answer _____

EXERCISE 3C

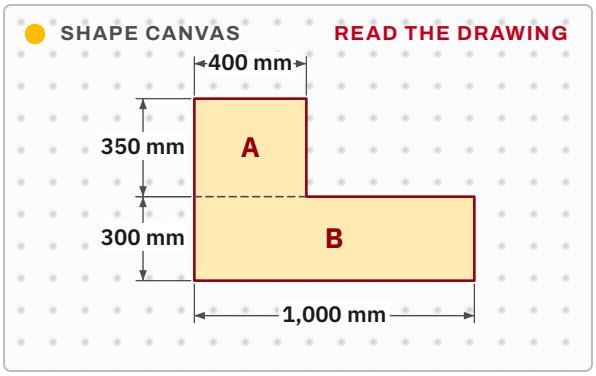
# Shape Canvas

Write the calculation for each drawing, then work it out

Name _____

Date _____

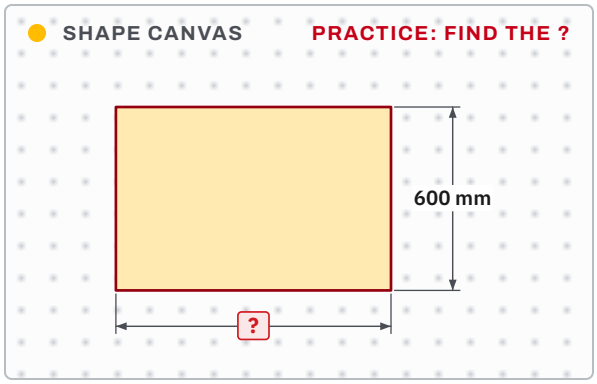
**1** The plate is split into A and B. Its area in mm² is 400 × 350 + 1,000 × 300. Work it out.




WORKING

Answer _____

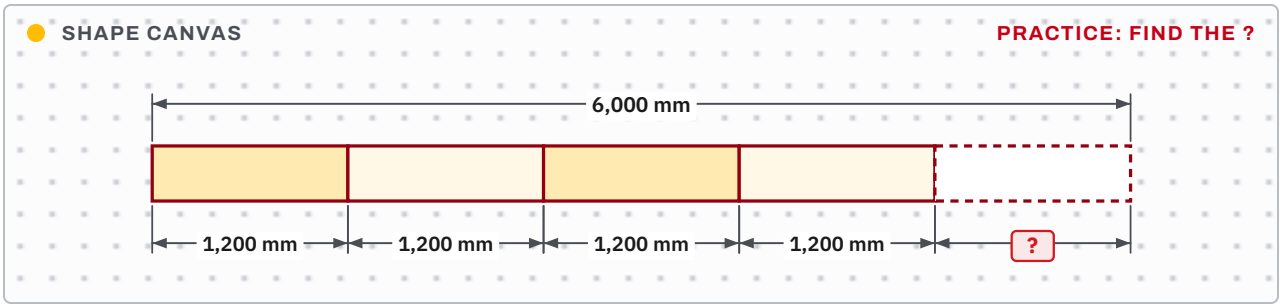
**2** The distance around this frame is 3,000 mm. Find the hidden length with L = 3,000 ÷ 2 − 600.




WORKING

Answer _____

**3** Four 1,200 mm pieces are cut from a 6,000 mm bar with four cuts. The kerf is 3 mm. The offcut is 6,000 − (4 × 1,200 + 4 × 3). Find it.

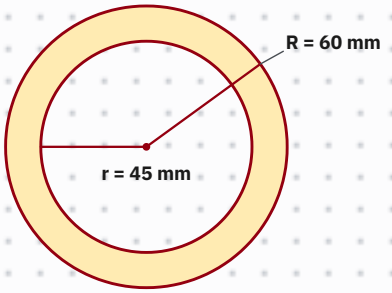



WORKING

Answer _____

**4** The ring has R = 60 mm and r = 45 mm. Its area is π × (R² − r²). Find it to two decimal places. Then change one value: what is the area if r becomes 30 mm?

SHAPE CANVAS — READ THE DRAWING

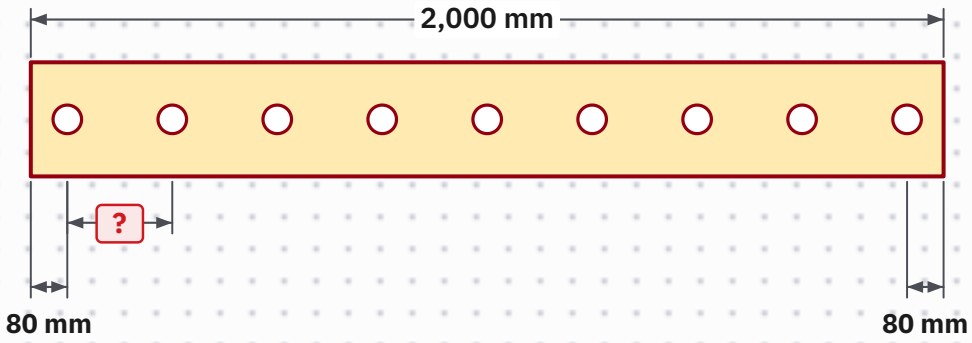



WORKING

Answer _____

**5** Nine holes are evenly spaced on a 2,000 mm plate. The spacing is (2,000 − 2 × 80) ÷ 8. Find it.

SHAPE CANVAS — PRACTICE: FIND THE ?




WORKING

Answer _____

EXERCISE 3D

# Positive and negative numbers

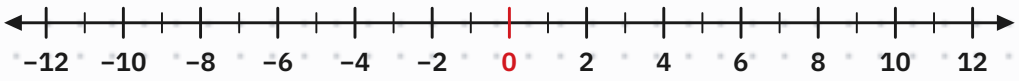
The number line, signs and the sign rules

Name _____

Date _____

**1** Put these numbers in order from least to greatest: 4, −7, 0, −2, 9, −11. Mark them on the number line first.

SHAPE CANVAS — MARK THE NUMBERS




WORKING

Answer _____

**2** −15 + 9

WORKING

Answer _____

**3** −6 − 9

WORKING

Answer _____

**4** 12 − (−7)

WORKING

Answer _____

**5** (−9) × 6

WORKING

Answer _____

**6** (−8) × (−7)

WORKING

Answer _____

**7** 72 ÷ (−9)

WORKING

Answer _____

**8** Find $(-4)^2$ and $-4^2$. Why are they different?

WORKING

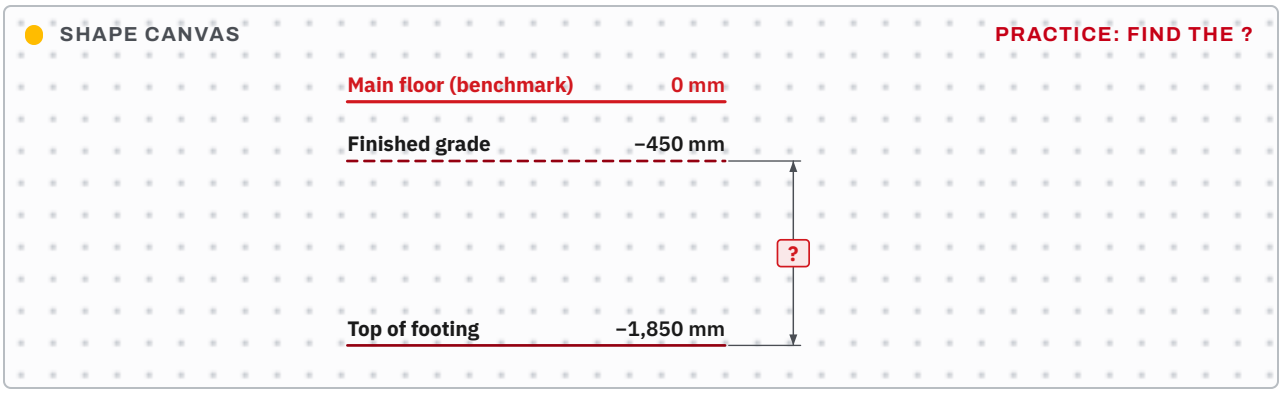
Answer _____

**9** At 7:00 it is −11 °C. By 14:00 the temperature has risen 15 degrees. By 20:00 it has fallen 8 degrees from the 14:00 reading. Find both temperatures.

WORKING

Answer _____

**10** On a site drawing the benchmark is the main floor, 0 mm. Finished grade is at −450 mm and the top of the footing is at −1,850 mm. How far is it from the top of the footing up to finished grade?




WORKING

Answer _____

**11** Six blocks should each be 914 mm long. They measure 916, 912, 913, 911, 915, 912 mm. Write each one as over (+) or under (−) the target. Find the total difference and the average difference, to one decimal place.

WORKING

Answer _____

**12** The temperature falls from 6 °C to −12 °C over 9 hours. Find the average change per hour.

WORKING

Answer _____

**13** A cooler at 2 °C drops 3 degrees each hour for 4 hours. Find its temperature with $2 + 4 \times (-3)$.

WORKING

Answer _____

EXERCISE 3K

# Carpentry and the BCBC

Stud counts and stair totals with BEDMAS

Name _____

Date _____

**1** Studs for a 7,200 mm wall at 400 mm o.c.

WORKING

Answer _____

**2** Studs for a 3,000 mm wall at 600 mm o.c.

WORKING

Answer _____

**3** Studs for a 20 ft wall at 16 in o.c.

WORKING

Answer _____

**4** Studs for a 5,300 mm wall at 400 mm o.c.

WORKING

Answer _____

**5** A wall is 14′ 6″ long with studs at 24 in o.c. Write one calculation for the number of spaces, then find the number of studs.

WORKING

Answer _____

**6** An exterior wall of 38 × 89 studs carries a roof plus one floor. The plans show the studs at 600 mm o.c. Table 9.23.10.1 of the BC Building Code (Sentence 9.23.10.1.(1), page 9-182) lists a maximum spacing of 400 mm for this wall. Is 600 mm allowed? How many studs does a 4,800 mm wall need at 400 mm o.c.?

WORKING

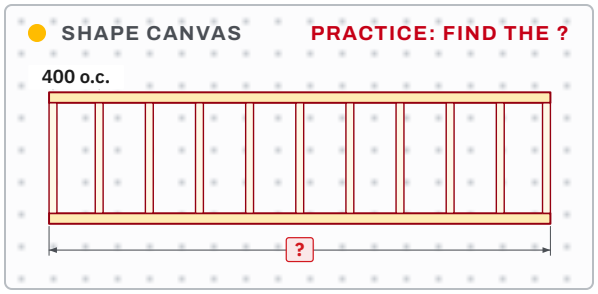
Answer _____

**7** For a 6,000 mm wall at 400 mm o.c., a learner keys 6,000 ÷ (400 + 1) instead of 6,000 ÷ 400 + 1. Find both answers to two decimal places and say which is the stud count.

WORKING

Answer _____

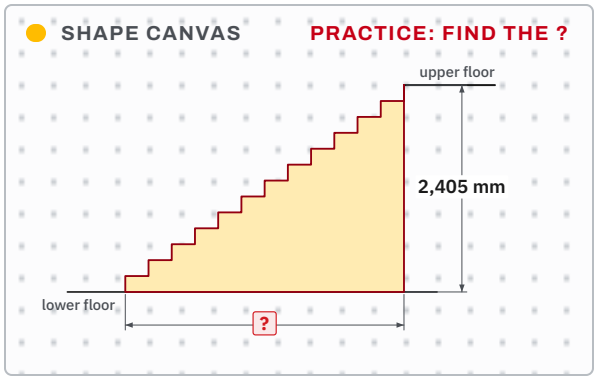
**8** Count the studs in the drawing. The spacing is 400 mm o.c. Find the wall length with (studs − 1) × spacing.




WORKING

Answer _____

**9** Find the total run of this stair with (risers − 1) × unit run. It has 13 risers of 185 mm and a unit run of 270 mm.




WORKING

Answer _____

**10** Check the total rise in the stair drawing above: 13 risers of 185 mm.

WORKING

Answer _____

**11** A total rise of 2,604 mm is split into 14 equal risers. Find the unit rise. Table 9.8.4.1 (Sentence 9.8.4.1. (1), page 9-30) allows a rise of 125 mm to 200 mm on a private stair. Is it allowed?

WORKING

Answer _____

**12** A private stair has a total rise of 2,960 mm. Find the unit rise with 14 risers and with 15 risers, to two decimal places. Which one meets the 200 mm maximum of Table 9.8.4.1?

WORKING

Answer _____

**13** One flight has 20 risers of 188 mm. Find its height. Article 9.8.3.3, Sentence (1) (page 9-30), says the vertical height of any flight shall not exceed 3.7 m. Does this flight meet it?

WORKING

Answer _____

# 4 Fractions

EXERCISES

Practice reading tapes and rules, writing fractions in lowest terms, changing between mixed numbers and improper fractions, and adding, subtracting, multiplying and dividing fractions. Give every answer in lowest terms.

**EXERCISE 4A**

## Skills

One skill at a time, from reading a rule to dividing fractions

Name _____

Date _____

**1** What fraction of the bar is shaded? Is it a proper fraction?

SHAPE CANVAS READ THE DRAWING

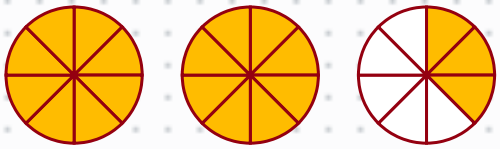


WORKING

Answer _____

**2** What fraction of the circles is shaded? Write it as an improper fraction and as a mixed number.

SHAPE CANVAS READ THE DRAWING

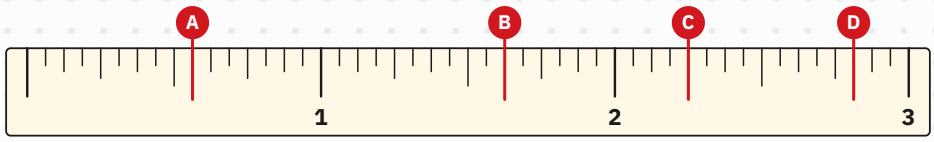


WORKING

Answer _____

**3** Read marks A, B, C and D to the nearest $\frac{1}{16}$ in.

SHAPE CANVAS READ THE DRAWING

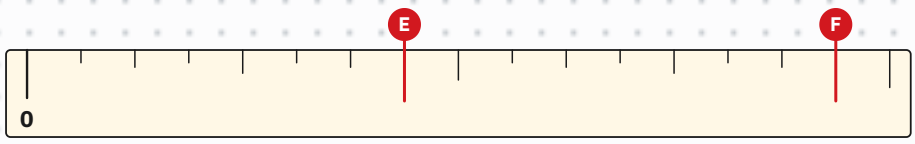



WORKING

Answer _____

**4** This half inch of a rule is marked in 32nds. Read marks E and F.

SHAPE CANVAS READ THE DRAWING

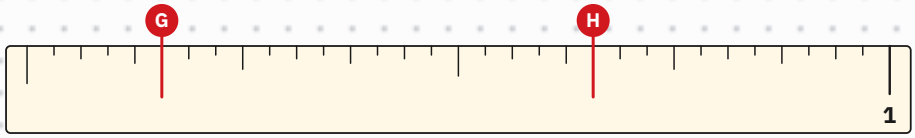



WORKING

Answer _____

**5** This part of a rule runs from $\frac{1}{2}$ in to 1 in and is marked in 64ths. Read marks G and H.

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

**6** Write $\frac{14}{32}$ and $\frac{36}{64}$ in lowest terms.

WORKING

Answer _____

**7** Write $\frac{7}{8}$ in 32nds and $\frac{5}{16}$ in 64ths.

WORKING

Answer _____

**8** Which is longer, $\frac{9}{16}$ in or $\frac{5}{8}$ in? By how much?

WORKING

Answer _____

**9** Put in order from smallest: $\frac{13}{32}$, $\frac{3}{8}$, $\frac{7}{16}$.

WORKING

Answer _____

**10** Change $\frac{37}{8}$ and $\frac{53}{16}$ to mixed numbers.

WORKING

Answer _____

**11** Change $4\frac{7}{16}$ and $6\frac{3}{4}$ to improper fractions.

WORKING

Answer _____

**12** The Sharp shows 3 ┌9 ┌16. Write it as a mixed number. If the TI-30Xa shows the same number (3⌟9⌟16), what does it show after 2nd d/c?

WORKING

Answer _____

**13** What is the reciprocal of $2\frac{1}{2}$?

WORKING

Answer _____

**14** Add. (a) $\frac{5}{16} + \frac{3}{4}$ (b) $7\frac{3}{8} + 2\frac{15}{16}$

WORKING

Answer _____

**15** Subtract. (a) $5 - 1\frac{11}{16}$ (b) $9\frac{1}{8} - 4\frac{3}{4}$

WORKING

Answer _____

**16** Multiply. (a) $\frac{5}{8} \times 12$ (b) $2\frac{1}{2} \times 1\frac{3}{4}$

WORKING

Answer _____

**17** Divide. (a) $\frac{3}{4} \div \frac{3}{16}$ (b) $15\frac{3}{4} \div 3$

WORKING

Answer _____

EXERCISE 4B

# Shop problems

Fractions in fabrication, pipe trades and millwright work

Name _____

Date _____

**1** A fabricator cuts pieces of $27\frac{5}{8}$ in, $19\frac{3}{16}$ in, $33\frac{1}{2}$ in from a 96 in bar. There are 3 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?

WORKING

Answer _____

**2** A pipe fitter needs two pieces of pipe, $18\frac{3}{4}$ in and $22\frac{7}{16}$ in long. What total length of pipe is that, ignoring the saw cuts?

WORKING

Answer _____

**3** A bolt passes through a $\frac{5}{8}$ in plate, a $\frac{3}{4}$ in plate and a $\frac{1}{8}$ in washer. What is the total thickness it clamps?

WORKING

Answer _____

**4** A millwright needs to fill a $\frac{3}{16}$ in gap with $\frac{1}{32}$ in shims. How many shims does it take?

WORKING

Answer _____

**5** A plate is $15\frac{5}{8}$ in wide. How far from the edge is the centre line?

WORKING

Answer _____

**6** Nine holes are evenly spaced along a 48 in angle. The end holes are $1\frac{3}{4}$ in from the ends. Find the spacing, centre to centre.

WORKING

Answer _____

**7** How many $4\frac{3}{8}$ in spacers can be cut from a 70 in bar? You make one cut for each spacer and the kerf is $\frac{1}{8}$ in. How long is the offcut?

WORKING

Answer _____

**8** A pipe has an outside diameter of $2\frac{3}{8}$ in and a wall $\frac{3}{16}$ in thick. What is the inside diameter?

WORKING

Answer _____

**9** A keyway measures $1\frac{11}{16}$ in wide on the rule. Will a $1\frac{3}{4}$ in key fit? By how much is it too wide or too narrow?

WORKING

Answer _____

**10** A 2,400 mm length of channel is cut so one piece is $\frac{3}{4}$ of the length. How long is that piece?

WORKING

Answer _____

**11** A 45 in bar is cut so one piece is $\frac{2}{3}$ of the bar. How long are the two pieces? Ignore the saw cut.

WORKING

Answer _____

**12** A 50 in bar is cut into 3 equal pieces. Ignore the saw cuts. How long is each piece, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

EXERCISE 4C

# Shape Canvas

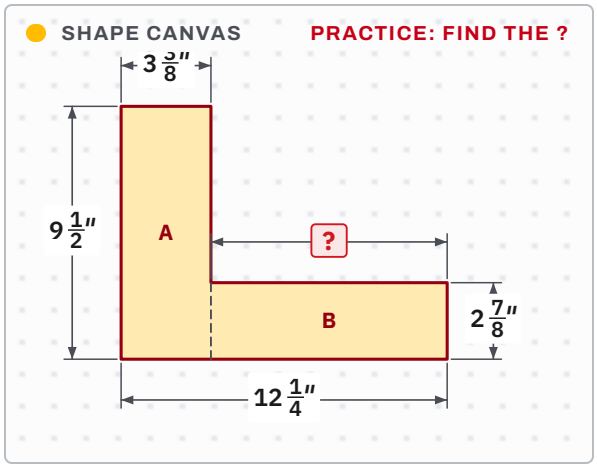
Parts in inches. Find every red ?

Name _____

Date _____

Each drawing is to scale. Break the part into pieces you know, and show the fractions you add or take away.

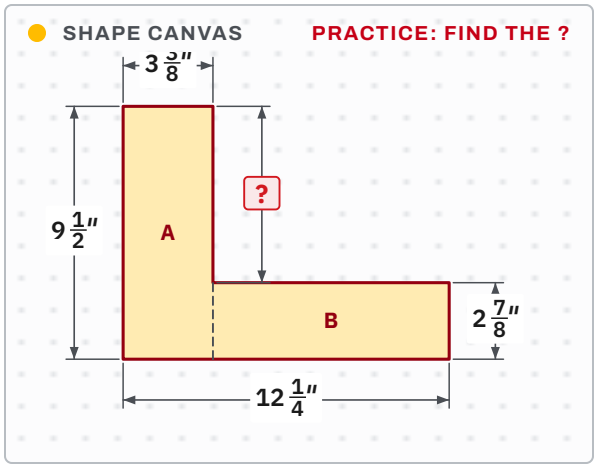
**1** Find the hidden width.




WORKING

Answer _____

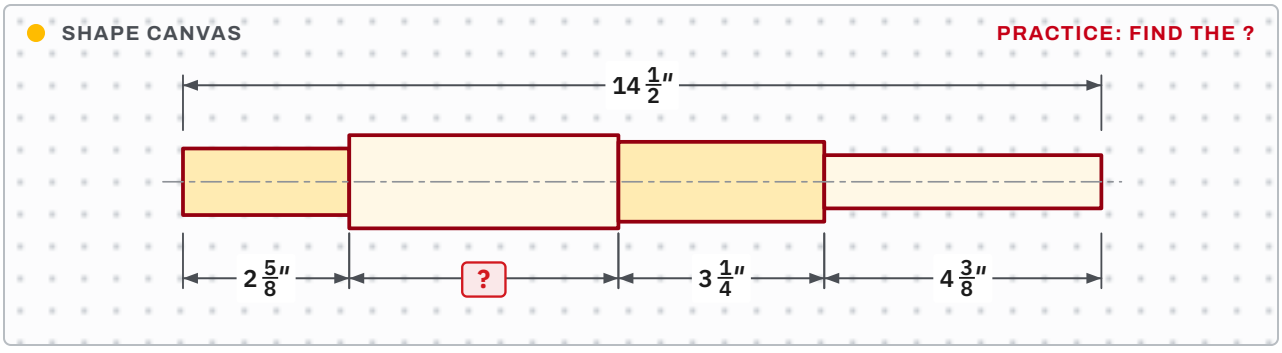
**2** Find the hidden height.




WORKING

Answer _____

**3** The shaft is $14\frac{1}{2}$ in long overall. Find the missing step.

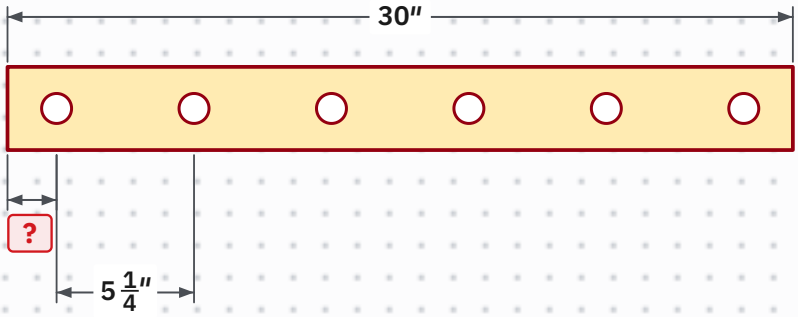



WORKING

Answer _____

**4** Six holes are evenly spaced along a 30 in plate. Find the distance from each end to the first hole.

SHAPE CANVAS — PRACTICE: FIND THE ?

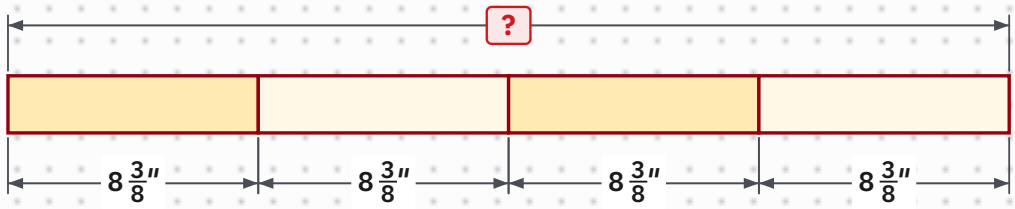



WORKING

Answer _____

**5** Change one value: four equal pieces of $8\frac{3}{8}$ in make up the bar. Find the overall length. If each piece is made $\frac{1}{16}$ in longer, what is the new overall length?

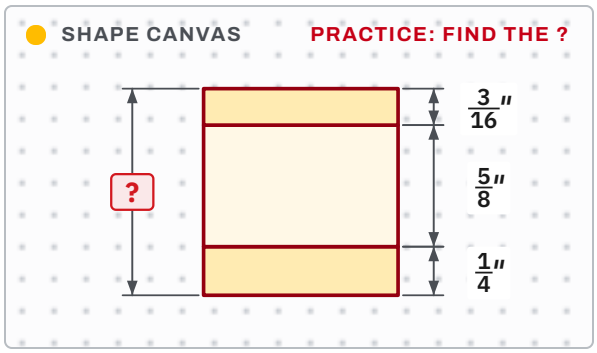
SHAPE CANVAS — PRACTICE: FIND THE ?




WORKING

Answer _____

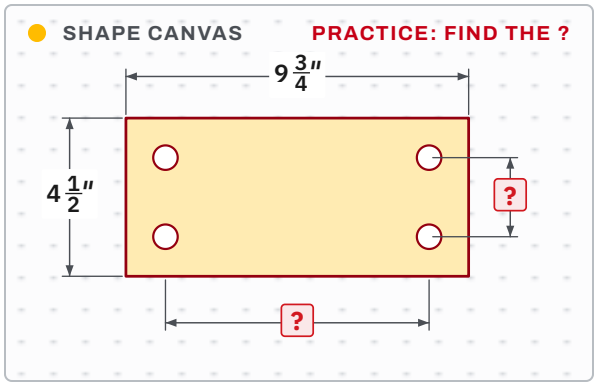
**6** Three plates are stacked. How thick is the stack?




WORKING

Answer _____

**7** Each hole centre is $1\frac{1}{8}$ in from two edges. Find the centre-to-centre distance along the plate and across it.




WORKING

Answer _____

**EXERCISE 4D**

## By hand

Add, subtract, multiply and divide fractions on paper. No calculator.

Name _____

Date _____

Find a common denominator to add or subtract. Change mixed numbers to improper fractions to multiply or divide. Give every answer in lowest terms.

**1** $\frac{3}{8} + \frac{5}{16}$

WORKING

Answer _____

**2** $2\frac{3}{4} + 1\frac{7}{8}$

WORKING

Answer _____

**3** $5\frac{1}{16} - 2\frac{3}{8}$

WORKING

Answer _____

**4** $7 - 3\frac{5}{16}$

WORKING

Answer _____

**5** $\frac{3}{4} \times \frac{2}{3}$

WORKING

Answer _____

**6** $2\frac{1}{2} \times 1\frac{3}{5}$

WORKING

Answer _____

**7** $4 \times 3\frac{5}{8}$

WORKING

Answer _____

**8** $\frac{5}{8} \div \frac{1}{4}$

WORKING

Answer _____

**9** $10\frac{1}{2} \div 1\frac{3}{4}$

WORKING

Answer _____

**10** $6\frac{3}{4} \div 3$

WORKING

Answer _____

EXERCISE 4K

# Carpentry and the BCBC

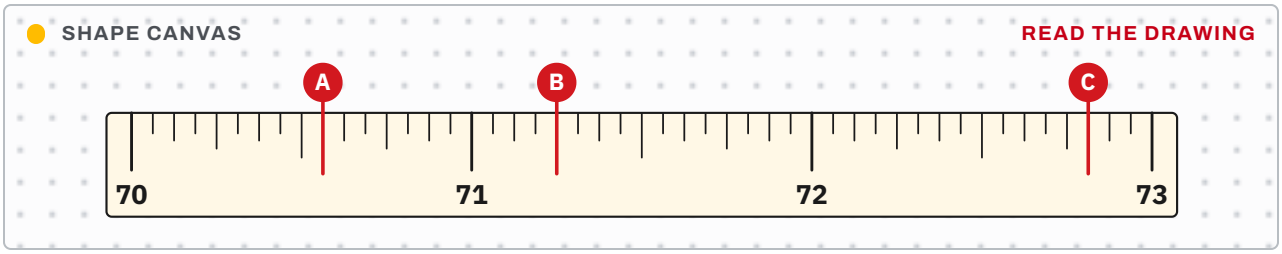
Tape readings on boards, cut lists with kerf, and code limits as fractions

Name _____

Date _____

Give tape readings in lowest terms. Every saw cut takes a $\frac{1}{8}$ in kerf unless the problem says otherwise. Code values are from BC Building Code 2024, Division B, in millimetres.

**1** Read marks A, B and C on the tape.




WORKING

Answer _____

**2** Pieces of $36\frac{3}{8}$ in, $42\frac{11}{16}$ in, $27\frac{1}{2}$ in are cut from a 120 in board with 3 cuts. How long is the offcut?

WORKING

Answer _____

**3** How many $22\frac{1}{2}$ in blocks come from a 96 in board, with one cut for each block? How long is the offcut?

WORKING

Answer _____

**4** A stud $92\frac{5}{8}$ in long is cut from a 96 in board with one cut. How much is left after the cut?

WORKING

Answer _____

**5** Find the centre of a wall $145\frac{3}{8}$ in long.

WORKING

Answer _____

**6** In a socket set marked in sixteenths, what size comes after $\frac{3}{4}$ in? Which is bigger, $\frac{11}{16}$ in or $\frac{5}{8}$ in, and by how much?

WORKING

Answer _____

**7** Article 9.23.5.1, Sentence (1) (page 9-175): holes in floor framing shall be not larger than one-quarter the depth of the member. What is the largest hole in a joist 140 mm deep?

WORKING

Answer _____

**8** Article 9.23.5.2, Sentence (1) (page 9-176): a notch is not deeper than one-third the joist depth and sits within half the joist depth from the edge of bearing. For a joist 235 mm deep, find both limits.

WORKING

Answer _____

**9** Article 9.23.5.3, Sentence (1) (page 9-176): the undamaged part of a stud shall not be less than two-thirds of its depth if it is loadbearing, or 40 mm if it is non-loadbearing. For an 89 mm deep stud, how deep can a notch be in each case?

WORKING

Answer _____

**10** Article 9.23.3.2, Sentence (1) (page 9-167): not less than half a nail's required length goes into the second member. A nail's required length is 63 mm. How far must it go into the second member?

WORKING

Answer _____

# 5 Decimals

EXERCISES

Practice place value, reading a metric rule, rounding, the four operations with decimals, changing between fractions and decimals, and money. Round only where the problem says so.

**EXERCISE 5A**

## Skills

Place value, rounding, the four operations and conversions

Name _____

Date _____

**1** What is the 4 worth in 30.249? Write 0.075 in words.

WORKING

Answer _____

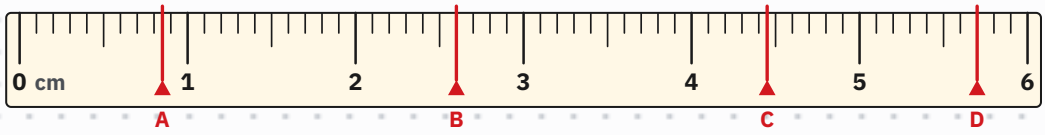
**2** Write $\frac{3}{100}$ and $\frac{48}{1{,}000}$ as decimals.

WORKING

Answer _____

**3** The rule is numbered in centimetres. Read marks A, B, C and D in millimetres, to the nearest 0.5 mm.

SHAPE CANVAS — READ THE DRAWING




WORKING

Answer _____

**4** Put in order from smallest: 0.4, 0.375, 0.41.

WORKING

Answer _____

**5** Which is longer, 2.5 mm or 2.45 mm? By how much?

WORKING

Answer _____

**6** Round 7.846 to one decimal place and to two decimal places.

WORKING

Answer _____

**7** Round 31.27 mm and 18.8 mm to the nearest 0.5 mm.

WORKING

Answer _____

**8** Add. 63.5 + 9.53 + 0.8

WORKING

Answer _____

**9** Subtract. 50 − 17.46

WORKING

Answer _____

**10** Multiply. (a) 0.6 × 0.4 (b) 25.4 × 7 (c) 3.125 × 1,000

WORKING

Answer _____

**11** Divide. (a) 8.4 ÷ 0.12 (b) 45.72 ÷ 6 (c) 19.5 ÷ 100

WORKING

Answer _____

**12** Change $\frac{7}{8}$ and $3\frac{1}{4}$ to decimals.

WORKING

Answer _____

**13** Change 0.875 and 0.08 to fractions in lowest terms.

WORKING

Answer _____

**14** Change 1.47 in to the nearest $\frac{1}{16}$ in and to the nearest $\frac{1}{32}$ in.

WORKING

Answer _____

**15** Change 0.29 in to the nearest $\frac{1}{16}$ in.

WORKING

Answer _____

**16** Change 30 mm to the nearest $\frac{1}{16}$ in.

WORKING

Answer _____

**17** Change $1\frac{1}{4}$ in to millimetres.

WORKING

Answer _____

EXERCISE 5B

# Shop problems

Decimals in fabrication, pipe trades and millwright work

Name _____

Date _____

**1** Pieces of 412.5 mm, 287.25 mm, 650 mm are cut from a 1,500 mm bar. There are 3 cuts and the kerf is 2.5 mm. How long is the offcut?

WORKING

Answer _____

**2** A pipe fitter needs 3 lengths of pipe, each 1.85 m. The pipe costs $14.60 per metre. What does the pipe cost?

WORKING

Answer _____

**3** A millwright stacks shims that are each 0.25 mm thick. How many shims make a stack 1.75 mm thick?

WORKING

Answer _____

**4** An order has 25 bolts at $0.46 each and 25 nuts at $0.12 each. What is the total?

WORKING

Answer _____

**5** Flat bar on a job weighs 2.36 kg per metre. What do 4.5 m weigh?

WORKING

Answer _____

**6** Eight holes are evenly spaced along a 900 mm plate. The end holes are 37.5 mm from the ends. Find the spacing to two decimal places.

WORKING

Answer _____

**7** A drawing gives a length of 3.38 in. What do you mark on the tape, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

**8** A metric part is 120 mm long. What is that to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

**9** A pre-apprentice works 37.5 hours at $21.40 an hour. What is the pay before deductions?

WORKING

Answer _____

**10** A $\frac{5}{8}$ in bolt hole is drilled in a metric part. What is the hole size in millimetres?

WORKING

Answer _____

EXERCISE 5C

# Shape Canvas

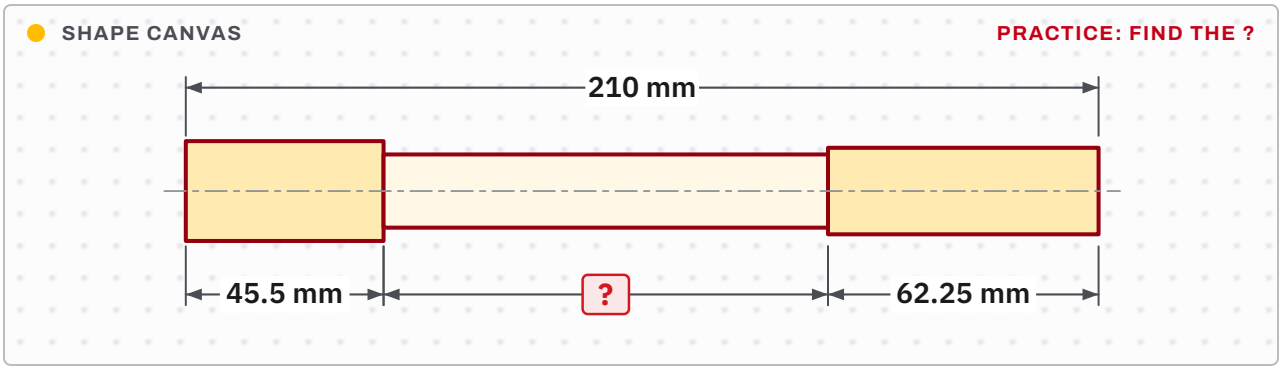
Decimal sizes on the canvas. Find every red ?

Name _____

Date _____

Each drawing is to scale. Line up the decimal points when you add or subtract.

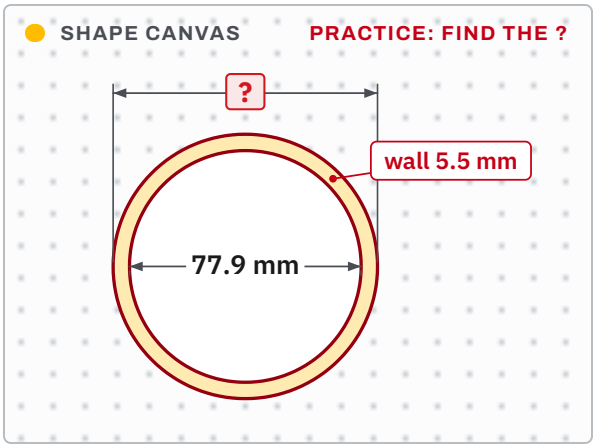
**1** The pin is 210 mm long overall. Find the length of the middle step.




WORKING

Answer _____

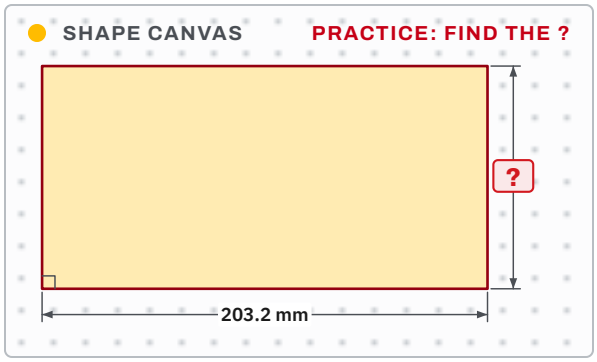
**2** Find the outside diameter of the pipe.




WORKING

Answer _____

**3** The distance around the plate is 609.6 mm. Find the width.

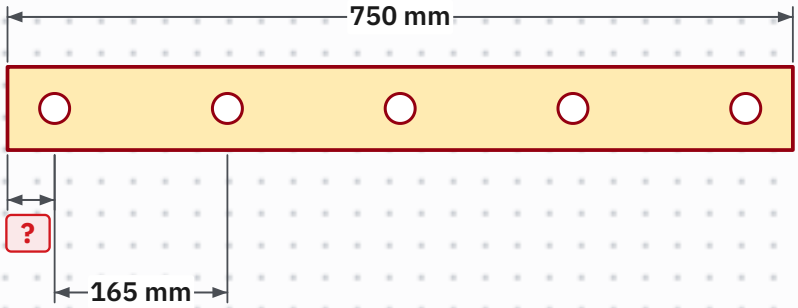



WORKING

Answer _____

**4** Five holes are evenly spaced along a 750 mm bar. Find the distance from each end to the first hole.

SHAPE CANVAS PRACTICE: FIND THE ?

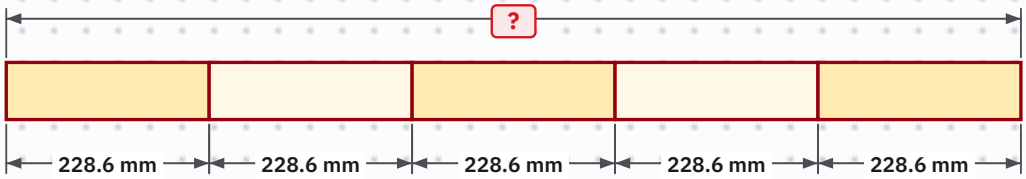



WORKING

Answer _____

**5** Change one value: five pieces of 228.6 mm make up the bar. Find the overall length. If each piece is cut 1.5 mm shorter, what is the new overall length?

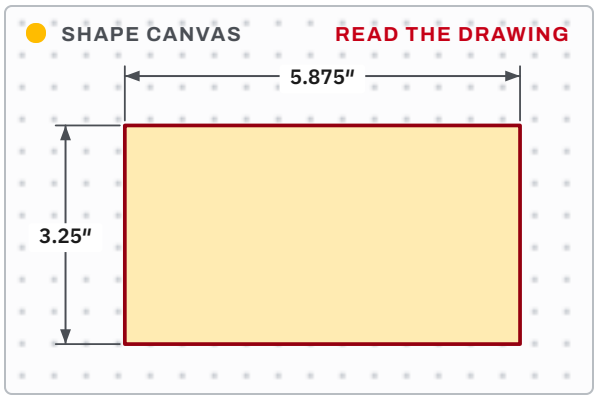
SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

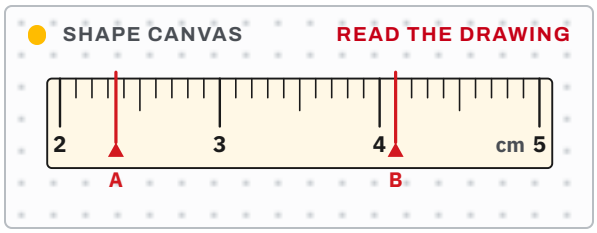
**6** Write each size as a fraction in lowest terms.




WORKING

Answer _____

**7** Read marks A and B to the nearest 0.5 mm. How far apart are they?




WORKING

Answer _____

**EXERCISE 5D**

## By hand

Decimal operations and rounding to hundredths and thousandths.
No calculator.

Name _____

Date _____

Line up the points to add or subtract. Count decimal places to multiply. To divide, keep going past the last digit by writing zeros, one place past where you round.

**1** 12.07 + 3.5 + 0.875

WORKING

Answer _____

**2** 20 – 7.384

WORKING

Answer _____

**3** 3.6 × 0.25

WORKING

Answer _____

**4** 14.75 × 2.4

WORKING

Answer _____

**5** 9.6 ÷ 0.4

WORKING

Answer _____

**6** 19.5 ÷ 6

WORKING

Answer _____

**7** 5 ÷ 8

WORKING

Answer _____

**8** 7 ÷ 3, to three decimal places

WORKING

Answer _____

**9** 2 ÷ 9, to two and to three decimal places

WORKING

Answer _____

**10** Change $\frac{3}{7}$ to a decimal, to three decimal places.

WORKING

Answer _____

EXERCISE 5K

# Carpentry and the BCBC

Decimal feet, lumber and sheet prices, and decimal code values

Name _____

Date _____

A foot has 12 inches. Give tape readings to the nearest 1/16 in and money to the nearest cent. Prices are given in each problem. Code values are from BC Building Code 2024, Division B.

**1** Change 9.75 ft to feet and inches.

WORKING

Answer _____

**2** Change 12 ft 4 in to decimal feet, to three decimal places.

WORKING

Answer _____

**3** A plan gives 7.85 ft. What is that in feet and inches, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

**4** A 25 ft board is cut into 7 equal pieces. Ignore the saw cuts. How long is each piece in decimal feet, to three decimal places, and in feet and inches, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

**5** Table 9.23.3.1 (Article 9.23.3.1, Sentence (2), page 9-166) gives minimum nail diameters: 2.87 mm for a 57 mm nail and 4.88 mm for a nail 101 mm or longer. How much bigger is the larger diameter?

WORKING

Answer _____

**6** An order has 18 pieces of 38 × 89 lumber, each 3.05 m long, at $2.65 per metre. Find the metres of lumber and the cost, to the nearest cent.

WORKING

Answer _____

**7** 14 sheets at $48.75 each and 3 boxes of nails at $27.99 each. Total before tax?

WORKING

Answer _____

**8** Trim runs around a room 4.27 m by 3.35 m. Ignore the door. Trim costs $2.19 per metre. What does it cost?

WORKING

Answer _____

**9** Table 9.23.10.1 (Article 9.23.10.1, Sentence (1), page 9-182) gives a maximum unsupported height of 3.0 m for 38 × 89 studs in an exterior wall that supports a roof only. A wall is 2.743 m tall. How far under the limit is it, in metres and in millimetres?

WORKING

Answer _____

# 6 Percent, Ratio and Proportion

EXERCISES

Practice with percents, decimals and fractions, percent of a number, allowances, increases and decreases, ratios, rates, proportions and scale drawings.

**EXERCISE 6A**

## Skills

One skill at a time

Name _____

Date _____

**1** Write 64% as a decimal.

WORKING

Answer _____

**2** Write 3.5% as a decimal.

WORKING

Answer _____

**3** Write 0.7 as a percent.

WORKING

Answer _____

**4** Write 2.4 as a percent.

WORKING

Answer _____

**5** Write 45% as a fraction in lowest terms.

WORKING

Answer _____

**6** Write $\frac{7}{8}$ as a percent.

WORKING

Answer _____

**7** Find 30% of 260.

WORKING

Answer _____

**8** Find 6.5% of 4,200.

WORKING

Answer _____

**9** 27 is what percent of 360?

WORKING

Answer _____

**10** 42 is 35% of what number?

WORKING

Answer _____

**11** Increase 340 by 15%.

WORKING

Answer _____

**12** Decrease 1,250 by 8%.

WORKING

Answer _____

**13** A value goes from 48 to 60. What is the percent increase?

WORKING

Answer _____

**14** Write 36 : 84 in lowest terms.

WORKING

Answer _____

**15** Write the ratio of 250 mm to 1.5 m in lowest terms.

WORKING

Answer _____

**16** Share 1,200 in the ratio 5 : 3.

WORKING

Answer _____

**17** A tank fills with 840 L in 12 minutes. What is the rate in L per minute?

WORKING

Answer _____

**18** Solve $\frac{7}{x} = \frac{21}{36}$.

WORKING

Answer _____

**19** At a scale of 1 : 40, a line measures 85 mm. How long is the real length?

WORKING

Answer _____

**20** A 14 mm part is drawn at 3 : 1. How long is it on the drawing?

WORKING

Answer _____

EXERCISE 6B

# Shop problems

Fabrication, pipe trades and millwright jobs

Name _____

Date _____

**1** A pipe run needs 64 m of pipe. Add a 12% waste allowance. The pipe on this job comes in 6.4 m lengths. How many metres do you need, and how many lengths do you order?

WORKING

Answer _____

**2** A fabrication job needs 18 sheets of 4 ft × 8 ft plate. Add a 15% allowance and round up to whole sheets.

WORKING

Answer _____

**3** A bearing lists at $86.50. Your shop gets a 12% trade discount. What does one bearing cost? What do 6 cost? Round to the cent.

WORKING

Answer _____

**4** A conveyor belt moves 18 m in 45 s. What is its rate in m per s? At that rate, how long does a part take to travel 50 m?

WORKING

Answer _____

**5** Of 85 welds tested, 6 are rejected. What percent are rejected, to two decimal places?

WORKING

Answer _____

**6** Cutting 14 brackets uses 9.8 m of angle iron. How much angle iron do 35 brackets use?

WORKING

Answer _____

**7** A two-part epoxy is mixed 5 : 2 by volume. You need 1.4 L of mixed epoxy. How much of each part do you measure?

WORKING

Answer _____

**8** A drawing is marked $\frac{1}{8}$″ = 1′-0″. A line measures $5\frac{3}{8}$ in. How long is the real length?

WORKING

Answer _____

**9** Pipe insulation goes from \$3.20 to \$3.68 per foot. What is the percent increase?

WORKING

Answer _____

**10** A drive pulley is 300 mm across and the driven pulley is 120 mm across. Write the ratio of the drive pulley to the driven pulley in lowest terms.

WORKING

Answer _____

**EXERCISE 6C**

# Shape Canvas

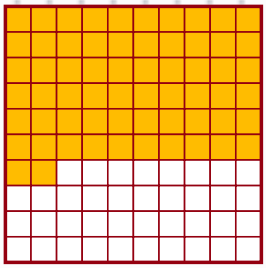
Find each red ?

Name _____

Date _____

**1** Write the shaded part as a percent, a decimal and a fraction in lowest terms.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

**2** The width is 35% of the length. Find the width.

SHAPE CANVAS PRACTICE: FIND THE ?

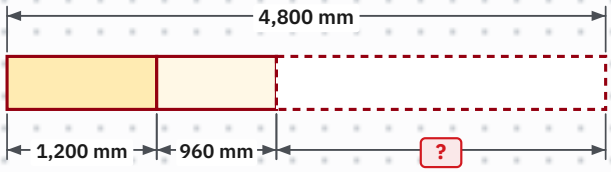



WORKING

Answer _____

**3** Two pieces are cut from a 4,800 mm bar. Find the offcut. What percent of the bar is the offcut?

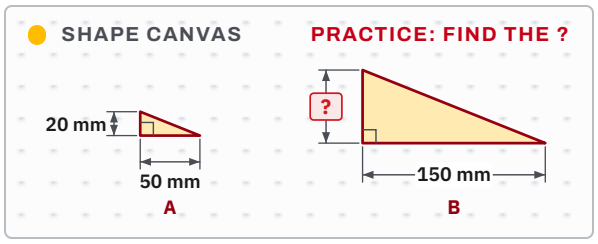
SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

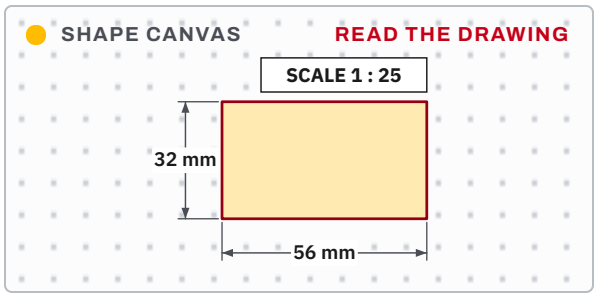
**4** Triangle B is triangle A made larger. Find the height of B.




WORKING

Answer _____

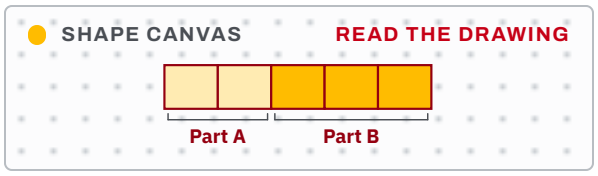
**5** The sizes are measured on a 1 : 25 drawing. Find the real length and width.




WORKING

Answer _____

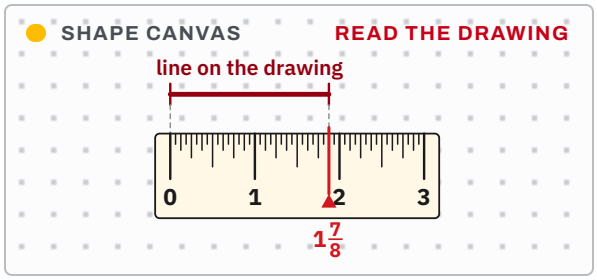
**6** Part A and Part B are mixed as shown. How much of each makes 1.5 L?




WORKING

Answer _____

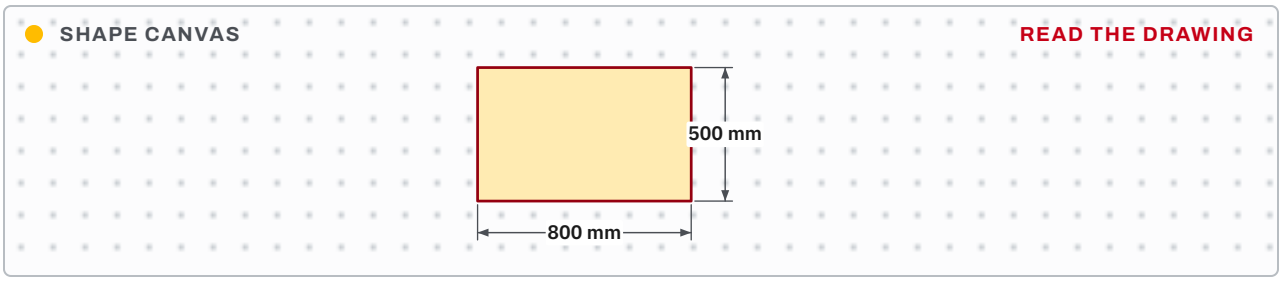
**7** A drawing is marked $\frac{1}{2}$″ = 1′-0″. The line above the rule is measured from 0 to the red mark. Find the real length of the line in feet and inches.




WORKING

Answer _____

**8** Change one value: a plate 800 mm × 500 mm is drawn at 1 : 10. Find the drawing sizes. Then find them at 1 : 5. What happens to every line?




WORKING

Answer _____

**EXERCISE 6D**

# By hand

No calculator: percents over 100%, conversions and quick percents

Name _____

Date _____

**1** Write 125% as a decimal.

WORKING

Answer _____

**2** Write 340% as a decimal.

WORKING

Answer _____

**3** Write 0.5% as a decimal.

WORKING

Answer _____

**4** Write 2.08 as a percent.

WORKING

Answer _____

**5** Write 0.015 as a percent.

WORKING

Answer _____

**6** Write $1\frac{1}{2}$ as a percent.

WORKING

Answer _____

**7** Write $2\frac{3}{5}$ as a percent.

WORKING

Answer _____

**8** Write $\frac{3}{20}$ as a percent.

WORKING

Answer _____

**9** Write 160% as a mixed number in lowest terms.

WORKING

Answer _____

**10** Write 225% as a mixed number in lowest terms.

WORKING

Answer _____

**11** Write 12.5% as a fraction in lowest terms.

WORKING

Answer _____

**12** Write $\frac{9}{25}$ as a decimal and as a percent.

WORKING

Answer _____

**13** Find 10% of 3,450.

WORKING

Answer _____

**14** Find 1% of 2,500.

WORKING

Answer _____

**15** Find 5% of 820. Hint: half of 10%.

WORKING

Answer _____

**16** Find 25% of 1,680. Hint: a quarter.

WORKING

Answer _____

**17** Find 35% of 240 as 10% + 10% + 10% + 5%.

WORKING

Answer _____

**18** Find 150% of 64 as 100% + 50%.

WORKING

Answer _____

**19** A new tank holds 140% of the old one. The old tank holds 500 L. What does the new one hold?

WORKING

Answer _____

**20** 18 is what percent of 12?

WORKING

Answer _____

EXERCISE 6E

# Money on the job

Pay, overtime, sales tax, discounts and unit price

Name _____

Date _____

Use GST 5% and B.C. PST 7%, the rates in Textbook Section 6.9. Find each tax on the price before tax, to the cent, then add.

**1** An apprentice works 38 hours at $24.50 an hour. What is the pay before deductions?

WORKING

Answer _____

**2** A worker earns $26.80 an hour for the first 40 hours. On this job, hours over 40 are paid at 1.5 times the hourly rate. This week the worker puts in 46 hours. Find the pay before deductions.

WORKING

Answer _____

**3** A pay stub shows gross pay of $1,240.00 and deductions of $287.15. What is the net pay?

WORKING

Answer _____

**4** A tool costs $185.00 before tax. Find the GST, the PST and the total.

WORKING

Answer _____

**5** A box of fittings costs $64.99 before tax. Find the GST, the PST and the total, to the cent.

WORKING

Answer _____

**6** A saw is listed at $420.00. It is on sale at 15% off. Take off the discount first, then add GST and PST. What do you pay?

WORKING

Answer _____

**7** Primer comes in a 3.78 L can for $44.98 and in a 0.946 L can for $15.49. Find the price per litre of each, to the cent. Which is the better buy?

WORKING

Answer _____

**8** A box of 100 screws costs $8.49. A box of 250 costs $19.75. Find the cost of one screw in each box, in cents to one decimal place. Which box costs less per screw?

WORKING

Answer _____

**9** Three apprentices each work 7.5 hours at $21.40 an hour. What is the total pay for the crew?

WORKING

Answer _____

**10** A service truck takes 46.2 L of fuel at $1.879 per litre, the price on the pump. What does the fill cost, to the cent?

WORKING

Answer _____

**11** A box of gloves goes from $12.50 to $13.75. What is the percent increase?

WORKING

Answer _____

**12** A shop charges $95.00 an hour for labour. A repair takes 2.75 hours and $48.60 in parts. Labour and parts are both taxable. Find the total with GST and PST.

WORKING

Answer _____

EXERCISE 6K

# Carpentry and the BCBC

Roof slope, waste allowance, scale drawings and crew time

Name _____

Date _____

Problems 1 and 2 use Table 9.26.3.1, Roofing Types and Slope Limits, which forms part of Sentence 9.26.3.1.(1) of the BC Building Code 2024 (BCBC 2024, Division B, Article 9.26.3.1. (page 9-241)). It gives a minimum slope of 1 in 3 for asphalt shingles in normal application and 1 in 6 in low slope application. Waste allowances and sheet sizes are given in each problem.

**1** A roof rises 5 in for every 12 in of run. Write the slope as "1 in ?". Is it at least the minimum slope for asphalt shingles in normal application (1 in 3, Table 9.26.3.1, page 9-242)?

WORKING

Answer _____

**2** A porch roof is framed at 3 in 12. Write it as "1 in ?". Can asphalt shingles go on in normal application (minimum 1 in 3)? In low slope application (minimum 1 in 6)? Use Table 9.26.3.1, page 9-242.

WORKING

Answer _____

**3** Write a slope of 1 in 3 as a rise in 12.

WORKING

Answer _____

**4** A roof is framed at 4 in 12. Its horizontal run is 3,600 mm. How far does the roof rise over that run?

WORKING

Answer _____

**5** A roof is framed at 6 in 12. Its run is 14 ft. How far does it rise, in feet?

WORKING

Answer _____

**6** A wall needs 42 sheets of sheathing. The estimate adds a 10% waste allowance. How many whole sheets do you order?

WORKING

Answer _____

**7** A floor is 36.5 m². The flooring estimate adds an 8% waste allowance. One box covers 2.2 m². How many m² do you need, and how many whole boxes?

WORKING

Answer _____

**8** On a floor plan drawn at 1 : 50, a wall measures 128 mm. How long is the real wall, in metres?

WORKING

Answer _____

**9** A 4,800 mm wall is drawn on a plan at 1 : 50. How long is it on the plan?

WORKING

Answer _____

**10** A plan is drawn at $\frac{1}{4}$″ = 1′-0″. A room measures $3\frac{1}{4}$ in by $2\frac{3}{4}$ in on the plan. What are the real sizes?

WORKING

Answer _____

**11** A concrete mix for a small pad is given as 1 part cement, 2 parts sand and 3 parts gravel by volume. How much of each is in 0.9 m³ of dry mix?

WORKING

Answer _____

**12** Four framers can sheathe a roof in 9 hours. How long would six framers take, working at the same rate?

WORKING

Answer _____

# 7 Measurement and Unit Conversion

EXERCISES

Practice with metric prefixes, feet and inches, exact conversion factors, area and volume units, litres and gallons, and temperature. Use 1 in = 25.4 mm, 1 ft = 0.3048 m, 1 lb = 0.453 592 37 kg, 1 US gal = 3.785 411 784 L and 1 imp gal = 4.546 09 L.

**EXERCISE 7A**

## Skills

One conversion at a time

Name _____

Date _____

**1** Convert 4.75 m to mm.

WORKING

Answer _____

**2** Convert 625 mm to m.

WORKING

Answer _____

**3** Convert 38 cm to mm.

WORKING

Answer _____

**4** Convert 3,250 mL to L.

WORKING

Answer _____

**5** Convert 2.4 t to kg.

WORKING

Answer _____

**6** Convert 750 g to kg.

WORKING

Answer _____

**7** Write 8′ 5″ in inches.

WORKING

Answer _____

**8** Write $9' \; 7\frac{3}{4}''$ in inches.

WORKING

Answer _____

**9** Write 94 in in feet and inches.

WORKING

Answer _____

**10** Write $157\frac{1}{2}$ in in feet and inches.

WORKING

Answer _____

**11** Convert $\frac{5}{8}$ in to mm.

WORKING

Answer _____

**12** Convert $2\frac{1}{4}$ in to mm.

WORKING

Answer _____

**13** Convert 950 mm to inches, to the nearest $\frac{1}{16}$ in.

WORKING

Answer _____

**14** Convert 15 ft to m.

WORKING

Answer _____

**15** Convert 75 lb to kg, to two decimal places.

WORKING

Answer _____

**16** How many feet are in 6 yd? How many metres?

WORKING

Answer _____

**17** Convert 2.5 $m^2$ to $mm^2$.

WORKING

Answer _____

**18** Convert 6 $ft^2$ to $in^2$.

WORKING

Answer _____

**19** Convert 25 $in^2$ to $mm^2$.

WORKING

Answer _____

**20** A tank is 800 mm × 400 mm × 300 mm inside. How many litres does it hold?

WORKING

Answer _____

**21** Convert 20 US gal to L, to two decimal places.

WORKING

Answer _____

**22** Convert 25 imp gal to L, to two decimal places.

WORKING

Answer _____

**23** Convert 35 °C to °F.

WORKING

Answer _____

**24** Convert −4 °F to °C.

WORKING

Answer _____

EXERCISE 7B

# Shop problems

Metric drawings, imperial tools and mixed units

Name _____

Date _____

**1** A metric drawing gives a length of 2,235 mm. Your tape reads in feet and inches. What do you mark, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

**2** Three pieces of pipe are $4' \, 6\frac{1}{2}''$, $7' \, 3\frac{3}{4}''$ and $2' \, 11''$ long. Find the total length in feet and inches and in millimetres.

WORKING

Answer _____

**3** 12 boxes of fittings weigh 22 lb each. What is the total mass in kilograms, to two decimal places?

WORKING

Answer _____

**4** A shop floor is 6.2 m by 4.5 m. Find its area in m² and in ft², to two decimal places. The coating label says one US gallon covers 350 ft². How many whole gallons do you buy?

WORKING

Answer _____

**5** A hydraulic reservoir is 900 mm × 600 mm × 450 mm inside. How many litres does it hold? How many US gallons, to two decimal places?

WORKING

Answer _____

**6** An older air receiver is marked 30 imp gal. What is that in US gallons, to two decimal places?

WORKING

Answer _____

**7** A welding procedure gives a preheat of 200 °F and a maximum temperature between passes of 120 °C. Write the preheat in °C, to one decimal place, and the maximum in °F.

WORKING

Answer _____

**8** A shaft warms up by 35 °C while running. How many °F is that rise?

WORKING

Answer _____

**9** A flat bar is $\frac{1}{2}$″ × 6″ × 30″. Find its volume in in³, then in cm³ to two decimal places.

WORKING

Answer _____

**10** A metric sheet is 1,220 mm × 2,440 mm. Find its area in m², then in ft² to two decimal places.

WORKING

Answer _____

**EXERCISE 7C**

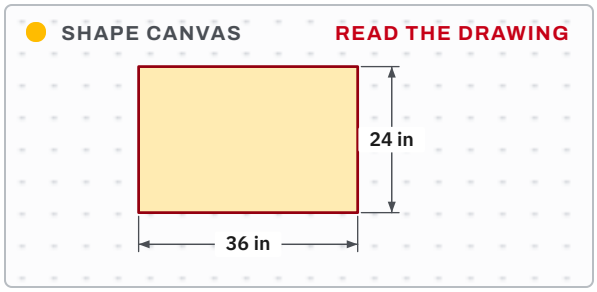
# Shape Canvas

Convert the drawings and find each red ?

Name _____

Date _____

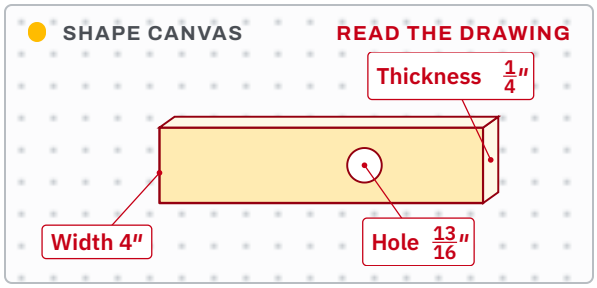
**1** Write both sizes of the sheet in millimetres.




WORKING

Answer _____

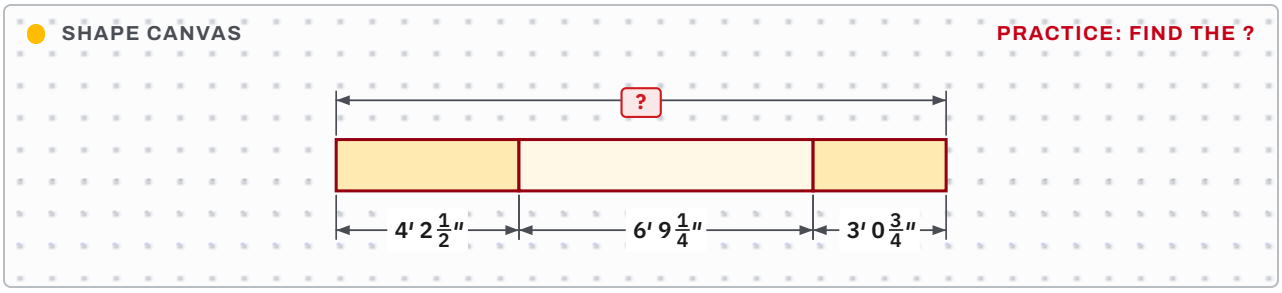
**2** Convert each size on the plate to millimetres.




WORKING

Answer _____

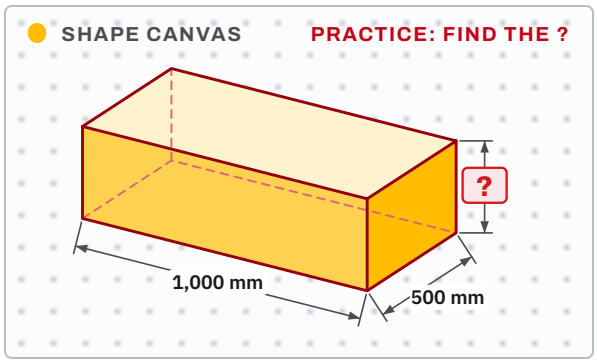
**3** Find the overall length in feet and inches and in millimetres.




WORKING

Answer _____

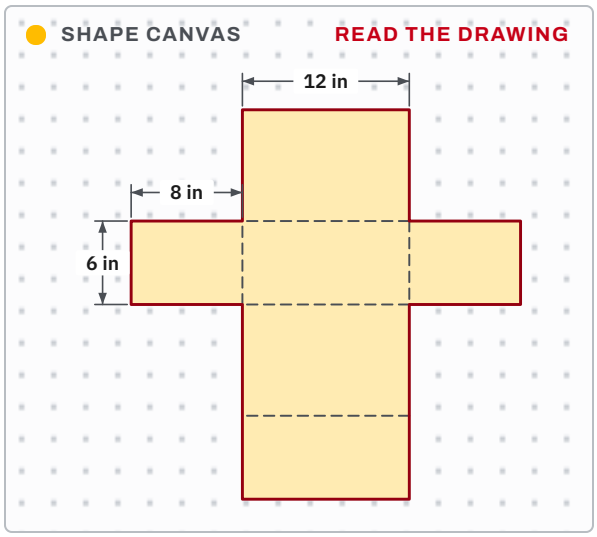
**4** The tank holds 150 L. Find the hidden height.




WORKING

Answer _____

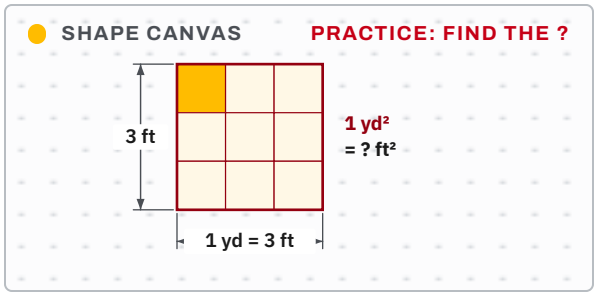
**5** Unfold it: the net of a closed box in inches. Find the total area in in² and in ft².




WORKING

Answer _____

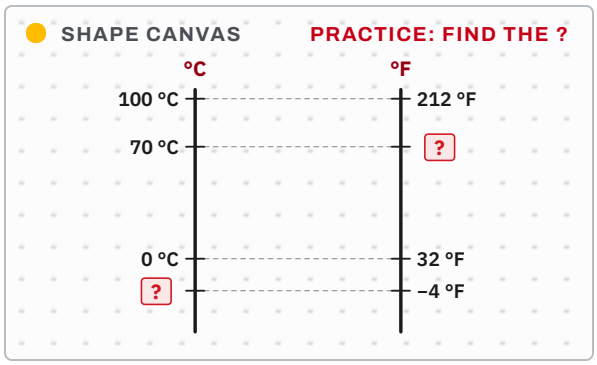
**6** A square yard split into square feet. How many ft² are in 1 yd²? How many in²?




WORKING

Answer _____

**7** Find each red ? on the two scales.




WORKING

Answer _____

**8** Change one value: a cube is 50 mm on each side. How many litres does it hold? Each side is doubled to 100 mm. How many litres now, and how many times as much?

WORKING

Answer _____

**EXERCISE 7D**

# Decimal feet, powers of ten and temperature

Textbook Sections 7.6, 7.7 and 7.8

Name _____

Date _____

Use 1 ft = 12 in, °F = 1.8 × °C + 32, °C = (°F − 32) ÷ 1.8, and 1.8 alone for a change in temperature. The imperial gallon is 4.546 09 L.

**1** Write 5′ 9″ in decimal feet.

WORKING

Answer _____

**2** Write 11′ 4″ in decimal feet, to two decimal places.

WORKING

Answer _____

**3** Write 7′ $10\frac{1}{2}$″ in decimal feet, to two decimal places.

WORKING

Answer _____

**4** Write 3′ $2\frac{5}{16}$″ in decimal feet, to two decimal places.

WORKING

Answer _____

**5** Write 8.5 ft in feet and inches.

WORKING

Answer _____

**6** Write 6.25 ft in feet and inches.

WORKING

Answer _____

**7** Write 10.8 ft in feet and inches, to the nearest $\frac{1}{16}$ in.

WORKING

Answer _____

**8** Write 23.47 ft in feet and inches, to the nearest $\frac{1}{16}$ in.

WORKING

Answer _____

**9** Write 52,000,000 in scientific notation.

WORKING

Answer _____

**10** Write 0.000 75 in scientific notation.

WORKING

Answer _____

**11** Write $9.1 \times 10^3$ as an ordinary number.

WORKING

Answer _____

**12** Work out $6 \times 10^5 \div (3 \times 10^2)$. Give the answer both ways.

WORKING

Answer _____

**13** Convert 15 °C to °F.

WORKING

Answer _____

**14** Convert 72 °F to °C, to one decimal place.

WORKING

Answer _____

**15** Convert −18 °C to °F.

WORKING

Answer _____

**16** Convert 350 °C to °F.

WORKING

Answer _____

**17** Convert 1,000 °F to °C, to one decimal place.

WORKING

Answer _____

**18** A temperature rises by 22 °C. How many °F is the rise?

WORKING

Answer _____

**19** A temperature drops by 27 °F. How many °C is the drop?

WORKING

Answer _____

**20** A 40 imp gal tank of water is heated from 10 °C to 60 °C. How many litres is the tank, to two decimal places? What is the rise in °F?

WORKING

Answer _____

EXERCISE 7K

# Carpentry and the BCBC

Code values in inches, board feet and floor area

Name _____

Date _____

Code values are quoted from the BC Building Code 2024, Division B, with the Sentence and page. The code value stays in millimetres. When you convert, round a maximum down and a minimum up to the nearest $\frac{1}{16}$ in. Board feet = thickness (in) × width (in) × length (ft) ÷ 12, and 1 board foot = 0.002 359 737 m³ (NIST SP 1038). 1 ft² = 0.092 903 04 m².

**1** Table 9.8.4.1 (Sentence 9.8.4.1.(1)), page 9-31, gives a maximum rise of 200 mm for private stairs. What is the largest tape reading, to a sixteenth, that stays within it?

WORKING

Answer _____

**2** The same table gives a minimum rise of 125 mm. What is the smallest tape reading, to a sixteenth, that meets it?

WORKING

Answer _____

**3** Table 9.8.4.2 (Sentence 9.8.4.2.(1), page 9-31) gives public stairs a minimum run of 280 mm. What is the smallest tape reading, to a sixteenth, that meets it?

WORKING

Answer _____

**4** Sentence 9.8.8.3.(1) (page 9-40) says guards shall be not less than 1 070 mm high, except as Sentences (2) and (3) allow. Convert 1 070 mm to inches, rounded up to a sixteenth.

WORKING

Answer _____

**5** Sentence 9.8.8.3.(3) (page 9-40) depends on whether a walking surface is more than 1 800 mm above the finished ground. A deck surface is measured at 5′ 11″ above the ground. Is it more than 1 800 mm? Show the height in millimetres.

WORKING

Answer _____

**6** Sentence 9.8.2.2.(2) (page 9-30) gives a minimum clear height over stairs of 2 050 mm. Write it in feet and inches, rounded up to a sixteenth.

WORKING

Answer _____

**7** How many board feet are in one piece 2 in × 12 in × 16 ft? In 18 pieces?

WORKING

Answer _____

**8** A bundle has 50 boards 1 in × 6 in × 8 ft. How many board feet is that?

WORKING

Answer _____

**9** Write 576 board feet in $m^3$, to three decimal places.

WORKING

Answer _____

**10** A floor is 5.4 m × 4.2 m. Find its area in $m^2$ and in $ft^2$, to two decimal places.

WORKING

Answer _____

**11** A room is 14 ft × 18 ft. Find its area in $ft^2$ and in $m^2$, to two decimal places.

WORKING

Answer _____

**12** Flooring for the 5.4 m × 4.2 m floor comes in boxes that cover 20 $ft^2$ each. How many whole boxes do you buy?

WORKING

Answer _____

# 8 Formulas and Transposing

EXERCISES

Practice reading formulas, substituting values, solving with inverse operations and transposing formulas, including formulas with squares and square roots. Check every answer by substituting it back.

**EXERCISE 8A**

## Skills

Name the parts, substitute, solve and transpose

Name _____

Date _____

**1** In $V = \pi \times r^2 \times h$, name the subject, the variables and the constant.

WORKING

Answer _____

**2** Rewrite $3ab$ and $\pi r^2$ with $\times$ signs.

WORKING

Answer _____

**3** Substitute: $A = l \times w$ with $l = 3.6$ m and $w = 2.5$ m.

WORKING

Answer _____

**4** Substitute: $P = 2 \times (l + w)$ with $l = 1{,}250$ mm and $w = 875$ mm.

WORKING

Answer _____

**5** Substitute: $C = \pi \times D$ with $D = 450$ mm. Round to two decimal places.

WORKING

Answer _____

**6** Substitute: $A = \pi \times r^2$ with $r = 35$ mm. Round to two decimal places.

WORKING

Answer _____

**7** Substitute: $V = l \times w \times h$ with $l = 1.2$ m, $w = 0.9$ m and $h = 0.75$ m.

WORKING

Answer _____

**8** Substitute: $t_F = 1.8 \times t_C + 32$ with $t_C = -10$ °C.

WORKING

Answer _____

**9** Solve $x + 175 = 640$.

WORKING

Answer _____

**10** Solve $x - 38 = 112$.

WORKING

Answer _____

**11** Solve $8 \times x = 2{,}000$.

WORKING

Answer _____

**12** Solve $x \div 6 = 45$.

WORKING

Answer _____

**13** Solve $15 \times x = 1{,}350$.

WORKING

Answer _____

**14** Solve $x - 2.75 = 9.5$.

WORKING

Answer _____

**15** Make l the subject of $A = l \times w$.

WORKING

Answer _____

**16** Make D the subject of $C = \pi \times D$.

WORKING

Answer _____

**17** Make w the subject of $V = l \times w \times h$.

WORKING

Answer _____

**18** Make s the subject of $P = 4 \times s$.

WORKING

Answer _____

**19** Make l the subject of $P = 2 \times (l + w)$.

WORKING

Answer _____

**20** Make $t_C$ the subject of $t_F = 1.8 \times t_C + 32$.

WORKING

Answer _____

**21** Make b the subject of $A = b \times h \div 2$.

WORKING

Answer _____

**22** Make h the subject of $T = a + r \times h$.

WORKING

Answer _____

**23** Make s the subject of $A = s^2$.

WORKING

Answer _____

**24** Make r the subject of $A = \pi \times r^2$.

WORKING

Answer _____

**25** Make h the subject of $V = \pi \times r^2 \times h$.

WORKING

Answer _____

**26** Make n the subject of $L = n \times \pi \times D$, where n is a number of turns.

WORKING

Answer _____

EXERCISE 8B

# Shop problems

Transpose first, then substitute and check

Name _____

Date _____

**1** A floor plate must cover 0.54 m² and is 0.6 m wide. How long is it?

WORKING

Answer _____

**2** A rectangular tank base is 1.5 m by 1.0 m. The tank must hold 1.8 m³. How tall is it?

WORKING

Answer _____

**3** A round gasket must have an area of 12,000 mm². Find its diameter to two decimal places.

WORKING

Answer _____

**4** A welded frame uses 6,400 mm of tube all the way around. It is 1,250 mm wide. How long is it?

WORKING

Answer _____

**5** A welding procedure calls for a preheat of 250 °F. Change it to °C, to two decimal places.

WORKING

Answer _____

**6** A square base plate has an area of 1,296 in². Find the side in inches and in feet.

WORKING

Answer _____

**7** A round tank has a radius of 0.75 m and must hold 4 m³. Use $V = \pi \times r^2 \times h$ to find its height, to two decimal places.

WORKING

Answer _____

**8** A rental invoice uses $T = 45 + 18.50 \times d$, with T in dollars and d in days. The total is $285.50. For how many days was the welder rented?

WORKING

Answer _____

**9** A tape wrapped once around a pipe reads $52\frac{3}{8}$ in. Find the diameter to the nearest 1/16 in.

WORKING

Answer _____

**10** A triangular gusset must have an area of 15,000 mm². Its height is 150 mm. Use $A = b \times h \div 2$ to find the base.

WORKING

Answer _____

**11** A drawing gives a disc an area of 0.25 m². Find its radius to three decimal places.

WORKING

Answer _____

**12** A shelf is 1.8 m long and 0.6 m deep, so its area is 1.08 m². A part laid flat is 1.6 m by 0.65 m. Find the area of the part. Does the part fit on the shelf? Explain using areas and sizes.

WORKING

Answer _____

**13** A sheet is 48 in wide and has an area of 4,608 in². Find its length in inches and in feet.

WORKING

Answer _____

**14** A cart wheel is 18 in in diameter. Use $n = L \div (\pi \times D)$ to find how many turns it makes when the cart moves 150 ft. Round to two decimal places.

WORKING

Answer _____

**15** Find the mistake. A learner transposes $P = 2 \times (l + w)$ to $l = P - 2 \times w$ and uses P = 4,000 mm and w = 700 mm. Find the learner's answer, check it by substituting back, and find the correct length.

WORKING

Answer _____

**EXERCISE 8C**

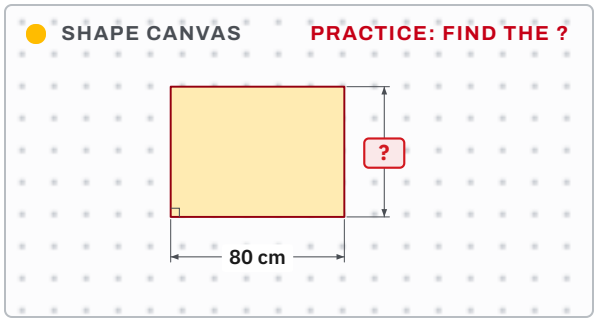
# Shape Canvas

Use the value you know to find the ?

Name _____

Date _____

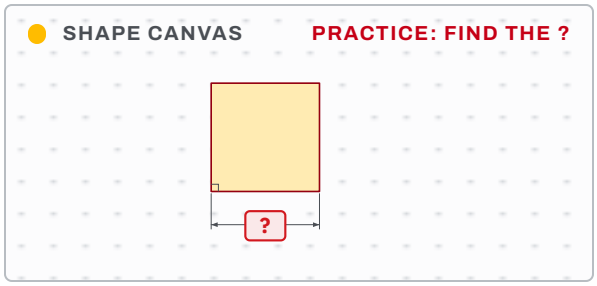
**1** The rectangle has an area of 4,800 cm². Find the width.




WORKING

Answer _____

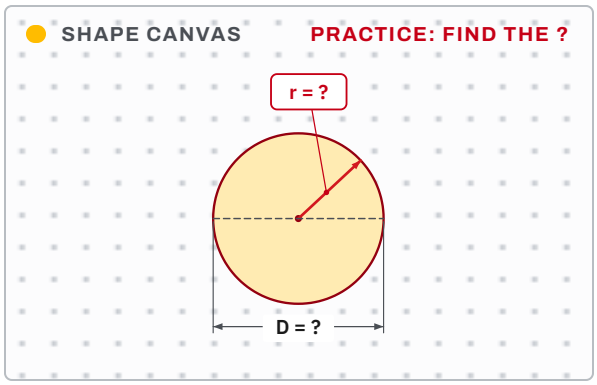
**2** The square has an area of 3,136 mm². Find the side.




WORKING

Answer _____

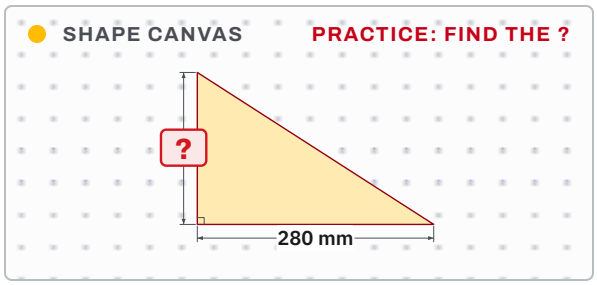
**3** The circle has an area of 8,000 mm². Find r and D to two decimal places.




WORKING

Answer _____

**4** The triangle has an area of 25,200 mm². Find the height.

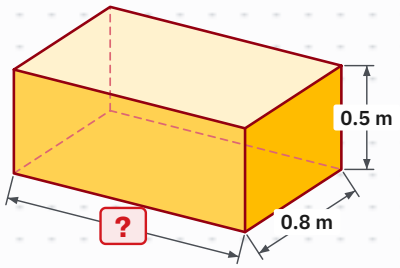



WORKING

Answer _____

**5** The box holds 0.48 m³. Find its length.

SHAPE CANVAS PRACTICE: FIND THE ?

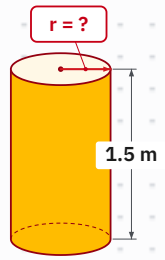



WORKING

Answer _____

**6** The tank holds 0.9 m³. Find its radius to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

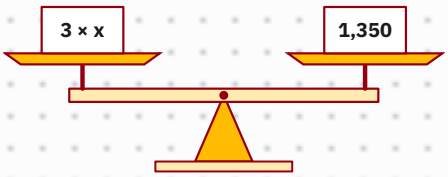



WORKING

Answer _____

**7** The scale is level. Find x.

SHAPE CANVAS READ THE DRAWING

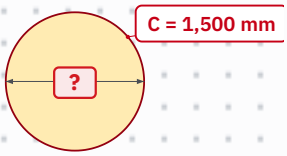



WORKING

Answer _____

**8** A tape around the pipe reads 1,500 mm. Find D to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

**9** Change one value: the square in problem 2 is given an area 9 times as big. Find the new side.

WORKING

Answer _____

**10** Change one value: the circle in problem 3 is given an area 4 times as big, 32,000 mm². Find the new diameter to two decimal places. What happened to it?

WORKING

Answer _____

EXERCISE 8K

# Carpentry and the BCBC

Stair formulas checked against BCBC 2024, Division B, Articles 9.8.3.3, 9.8.4.1, 9.8.4.2 and 9.8.4.4

Name _____

Date _____

**1** A private stair has a total rise of 2,565 mm. The maximum rise is 200 mm (Table 9.8.4.1, page 9-30). Find the number of risers and the unit rise to two decimal places.

WORKING

Answer _____

**2** The stair in problem 1 has a unit run of 265 mm. Does the run meet Table 9.8.4.2 (page 9-31) for a private stair? Find the total run.

WORKING

Answer _____

**3** A public stair has a total rise of 2,960 mm. The maximum rise is 180 mm (Table 9.8.4.1). Find the number of risers and the unit rise to two decimal places.

WORKING

Answer _____

**4** For the stair in problem 3, a learner rounds down to 16 risers. Find the unit rise and explain why it does not meet Table 9.8.4.1.

WORKING

Answer _____

**5** A stair has 16 risers of 178 mm. Find the total rise.

WORKING

Answer _____

**6** A private stair has 13 treads and a total run of 3,315 mm. Find the unit run. Does it meet Table 9.8.4.2?

WORKING

Answer _____

**7** The risers of a flight measure 182, 185, 181, 186, 184 and 179 mm, from the bottom. Do they meet Sentence 9.8.4.4.(1) (page 9-32): 5 mm between adjacent risers and 10 mm between the tallest and the shortest?

WORKING

Answer _____

**8** The rectangular treads of a flight have runs of 270, 276, 268 and 272 mm. Sentence 9.8.4.4.(3) allows 5 mm between adjacent treads. Do they meet it?

WORKING

Answer _____

**9** Article 9.8.3.3 (page 9-30) says the vertical height of a flight shall not exceed 3.7 m. A private stair has a total rise of 3,900 mm. Can it be one flight? If it is split into two equal flights with a landing, find the risers and the unit rise in each flight.

WORKING

Answer _____

**10** A stair has n risers of 190 mm and a total rise of 2,660 mm. Write an equation and solve it for n.

WORKING

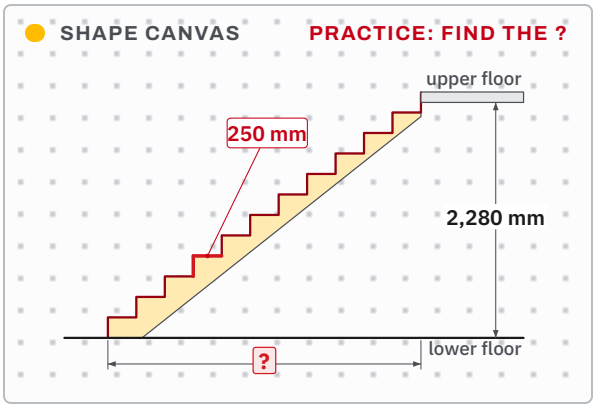
Answer _____

**11** A stair has 15 risers and a total run of 3,640 mm. Transpose
total run = number of treads × unit run and find the unit run.

WORKING

Answer _____

**12** The stair drawing shows 12 risers, a total rise of 2,280 mm and a unit run of 250 mm. Find the unit rise and the total run marked ?.




WORKING

Answer _____

**13** A private stair has a total rise of 2,600 mm, built as 13 risers of 200 mm. It must fit in a total run of no more than 3,400 mm. Find the largest whole-millimetre unit run that fits, and check it against Table 9.8.4.2.

WORKING

Answer _____

**14** Change one value: the total rise in problem 1 becomes 2,605 mm. Find the number of risers and the unit rise to two decimal places. What changed?

WORKING

Answer _____

# 9 Perimeter and Circumference

EXERCISES

Practice finding the distance around straight-sided shapes and circles, the stretch-out of pipes and ducts, arcs and composite shapes. The distance around a circle is always π × D.

**EXERCISE 9A**

## Skills

Perimeter, parts of a circle, circumference and arcs

Name _____

Date _____

**1** Find the perimeter of a rectangle 2,400 mm by 1,150 mm.

WORKING

Answer _____

**2** Find the perimeter of a square with sides of 75 cm.

WORKING

Answer _____

**3** A triangle has sides of 0.8 m, 1.1 m and 1.3 m. Find the perimeter.

WORKING

Answer _____

**4** A regular pentagon has sides of 90 mm. Find the perimeter.

WORKING

Answer _____

**5** A regular octagon has sides of 45 in. Find the perimeter in feet.

WORKING

Answer _____

**6** Find the perimeter of a rectangle $15\frac{3}{4}$ in by $9\frac{5}{8}$ in.

WORKING

Answer _____

**7** Find the perimeter of a rectangle 9 ft 3 in by 5 ft 10 in, in feet and inches.

WORKING

Answer _____

**8** A rectangle has a perimeter of 3,800 mm and a length of 1,200 mm. Find the width.

WORKING

Answer _____

**9** A circle has a diameter of 273 mm. Find the radius.

WORKING

Answer _____

**10** A circle has a radius of 82.5 mm. Find the diameter.

WORKING

Answer _____

**11** D = 380 mm. Find C to two decimal places.

WORKING

Answer _____

**12** r = 95 mm. Find C to two decimal places.

WORKING

Answer _____

**13** C = 1,500 mm. Find D to two decimal places.

WORKING

Answer _____

**14** D = 14 in. Find C to the nearest 1/16 in.

WORKING

Answer _____

**15** Find the length of a 45° arc of a circle with D = 800 mm. Round to two decimal places.

WORKING

Answer _____

**16** Find the length of a 150° arc of a circle with r = 600 mm. Round to two decimal places.

WORKING

Answer _____

**17** Find the curved edge of a half circle with D = 640 mm, to two decimal places.

WORKING

Answer _____

**18** A circle has its diameter cut in half. What happens to its circumference?

WORKING

Answer _____

EXERCISE 9B

# Shop problems

Fabrication, pipe trades and millwright

Name _____

Date _____

**1** A frame of angle iron is 2 ft 6 in by 1 ft 9 in. How much angle iron goes around it? Give feet and inches.

WORKING

Answer _____

**2** A steel band goes once around a tank 1,850 mm in diameter, with 150 mm extra for the overlap. Find the band length to the nearest millimetre.

WORKING

Answer _____

**3** A round duct is 355 mm in diameter. The drawing adds 25 mm to the stretch-out for the seam. Find the cut length to the nearest millimetre.

WORKING

Answer _____

**4** An insulation jacket wraps a pipe with an outside diameter of $10\frac{3}{4}$ in. Find the stretch-out to the nearest 1/16 in.

WORKING

Answer _____

**5** A pulley 250 mm in diameter turns 1,450 times. How far does a point on its rim travel, in metres to two decimal places?

WORKING

Answer _____

**6** A cart wheel is 24 in in diameter. How many turns does it make in 100 ft? Round to two decimal places.

WORKING

Answer _____

**7** A fillet weld runs once around a pipe with an outside diameter of 168.3 mm. How long is the weld, to two decimal places?

WORKING

Answer _____

**8** A tape wrapped around a round column reads 2,042 mm. Find the diameter to the nearest millimetre.

WORKING

Answer _____

**9** Four guard frames, each 1,200 mm by 800 mm, are cut from 6,000 mm lengths of flat bar. Ignore saw cuts. How much bar is needed, and how many 6,000 mm lengths must be ordered?

WORKING

Answer _____

**10** A handrail follows a 90° arc of a circle with a 1,500 mm radius. How long is the rail, to two decimal places?

WORKING

Answer _____

**11** A belt runs around two pulleys, each 300 mm in diameter, with centres 1,100 mm apart. The pulleys are moved 150 mm farther apart. Find the belt length before and after. How much longer is the new belt? Round to two decimal places.

WORKING

Answer _____

**12** A round tank lid is 1,220 mm in diameter. A rubber gasket goes once around its edge, with 40 mm extra for the joint. Find the gasket length to two decimal places.

WORKING

Answer _____

**13** A rectangular duct is 600 mm by 400 mm. A strip of insulation wraps once around it with a 50 mm overlap. How long must the strip be?

WORKING

Answer _____

EXERCISE 9C

# Shape Canvas

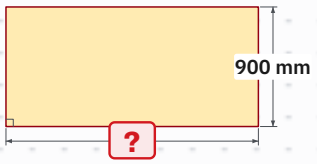
Find every ?, break it down, unfold it. Round answers to two decimal places.

Name _____

Date _____

**1** The perimeter is 5,600 mm. Find the length.

SHAPE CANVAS — PRACTICE: FIND THE ?

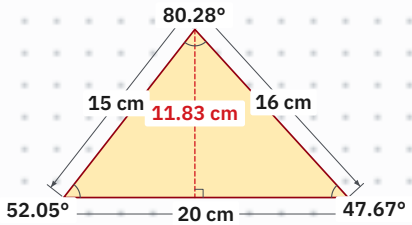



WORKING

Answer _____

**2** Find the perimeter. The dashed line is the height, not an edge.

SHAPE CANVAS — READ THE DRAWING

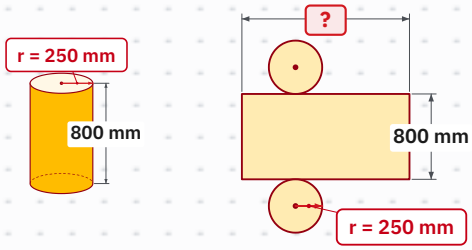



WORKING

Answer _____

**3** Unfold it. Find the stretch-out marked ?.

SHAPE CANVAS — PRACTICE: FIND THE ?

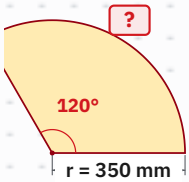



WORKING

Answer _____

**4** Find the length of the arc marked ?.

SHAPE CANVAS — PRACTICE: FIND THE ?

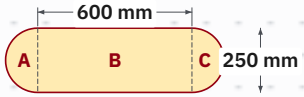



WORKING

Answer _____

**5** Break it down: the plate has a half circle on each end. Find its perimeter.

SHAPE CANVAS READ THE DRAWING

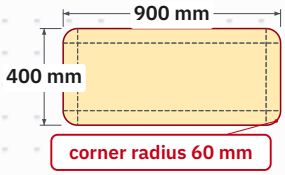



WORKING

Answer _____

**6** Each corner is a quarter circle. Find the perimeter of the plate.

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

**7** The belt is 2,400 mm long. Find the centre distance marked ?.

SHAPE CANVAS PRACTICE: FIND THE ?

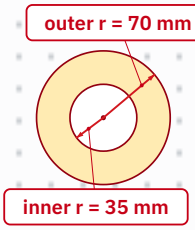



WORKING

Answer _____

**8** Change one value: the outside of the washer is kept, but the hole is made 10 mm bigger across. Find the new total length of both edges.

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

EXERCISE 9K

# Carpentry and the BCBC

Wall plates (BCBC 2024, Division B, Articles 9.23.11.2 and 9.23.11.3), rim joists, fascia and trim

Name _____

Date _____

**1** A loadbearing wall 7,320 mm long has one bottom plate and two top plates (Sentences 9.23.11.2.(1), page 9-184, and 9.23.11.3.(1), page 9-185). How much plate is in the wall?

WORKING

Answer _____

**2** Sentence 9.23.11.3.(3) permits a single top plate "in all non-loadbearing walls". A non-loadbearing wall 3,050 mm long has one bottom plate and one top plate. How much plate is that?

WORKING

Answer _____

**3** The outside walls of a house are 14,640 mm by 9,760 mm. All are loadbearing with one bottom and two top plates. Using the outside perimeter and ignoring openings, find the plate needed and the number of 4,880 mm lengths.

WORKING

Answer _____

**4** A floor frame is 10,980 mm by 7,930 mm. Find the total length of rim joist. How many 3,660 mm lengths are needed, ignoring joints?

WORKING

Answer _____

**5** Walls are 12,000 mm by 8,000 mm. The hip roof overhangs 600 mm on every side, with fascia on all four edges. Find the length of fascia.

WORKING

Answer _____

**6** Walls are 36 ft by 24 ft. The hip roof overhangs 18 in on every side. Find the length of fascia in feet.

WORKING

Answer _____

**7** A room is 4,270 mm by 3,660 mm. Baseboard runs around it except across a door opening 860 mm wide and a closet opening 1,520 mm wide. How much baseboard is needed?

WORKING

Answer _____

**8** A room is 13 ft 4 in by 10 ft 8 in with one door opening 30 in wide. How much baseboard is needed? Give feet and inches.

WORKING

Answer _____

**9** Window casing is measured around its outside edge, 1,370 mm by 1,060 mm. How much casing goes around the window?

WORKING

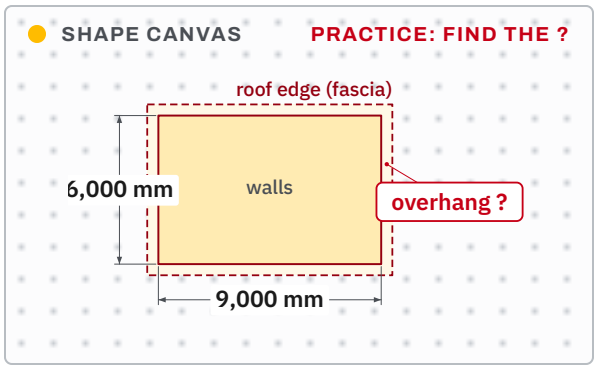
Answer _____

**10** Change one value: the house in problem 3 is made 1,220 mm longer. How much more plate is needed?

WORKING

Answer _____

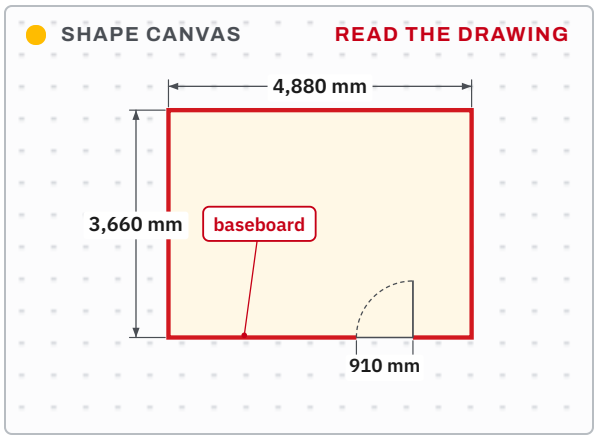
**11** The walls are 9,000 mm by 6,000 mm and the fascia around the roof is 33,600 mm long. Find the overhang marked ?.




WORKING

Answer _____

**12** Find the length of baseboard for the room. The door opening is not covered.




WORKING

Answer _____

# 10 Area

EXERCISES

Practice with square units, the area formulas for rectangles, triangles, parallelograms, trapezoids, circles and rings, composite shapes and surface area from nets. Use the π key for circles.

**EXERCISE 10A**

## Skills

One skill at a time

Name _____

Date _____

**1** A plate drawn on a 1 cm grid covers 5 rows of 9 squares. What is its area?

WORKING

Answer _____

**2** Write 1,500,000 mm² in m².

WORKING

Answer _____

**3** Write 0.08 m² in mm².

WORKING

Answer _____

**4** Write 6 ft² in in². Then write 432 in² in ft².

WORKING

Answer _____

**5** Find the area of a rectangle 840 mm by 350 mm.

WORKING

Answer _____

**6** Find the area of a square with 1.6 m sides.

WORKING

Answer _____

**7** Triangle: base 520 mm, height 270 mm. Find the area.

WORKING

Answer _____

**8** Triangle: base 15 in, height 7 in. Find the area.

WORKING

Answer _____

**9** Parallelogram: base 0.75 m, slanted side 0.5 m, height 0.4 m. Find the area.

WORKING

Answer _____

**10** Trapezoid: parallel sides 250 mm and 410 mm, height 180 mm. Find the area.

WORKING

Answer _____

**11** Circle, radius 45 mm. Find the area to two decimal places.

WORKING

Answer _____

**12** Circle, diameter 30 in. Find the area to two decimal places.

WORKING

Answer _____

**13** Half circle, radius 1.2 m. Find the area to two decimal places.

WORKING

Answer _____

**14** Quarter circle, radius 500 mm. Find the area to the nearest $mm^2$.

WORKING

Answer _____

**15** Ring: outside diameter 120 mm, inside diameter 80 mm. Find the area to two decimal places.

WORKING

Answer _____

**16** A ring has an outside radius of 60 mm and an inside radius of 40 mm. A student finds its area with $\pi \times (60 - 40)^2$. Explain the mistake.

WORKING

Answer _____

**17** A triangle has an area of 54 $in^2$ and a base of 12 in. Find its height.

WORKING

Answer _____

**18** Write 2.5 $m^2$ in $cm^2$. (1 $m^2$ = 10,000 $cm^2$)

WORKING

Answer _____

EXERCISE 10B

# Shop problems

Area at work, metric and imperial

Name _____

Date _____

**1** A mezzanine floor is 7.2 m by 3.6 m. It is covered with checker plate sheets that are 1.2 m by 2.4 m, laid edge to edge with no waste. Find the floor area and the number of sheets.

WORKING

Answer _____

**2** Four plates, each 2.5 m by 1.6 m, are primed on both sides. The primer label says 1 L covers 7 m². Primer is sold in 1 L tins. How many tins do you buy? Give the litres to two decimal places.

WORKING

Answer _____

**3** A full-face gasket is a ring with an outside diameter of 340 mm and an inside diameter of 220 mm. Find the area of gasket material to two decimal places.

WORKING

Answer _____

**4** A job needs 24 triangular gussets, each with a base of 16 in and a height of 12 in. Find the total area in in² and in ft².

WORKING

Answer _____

**5** A base plate is 450 mm square with four 22 mm diameter bolt holes. Find the area of steel left, to two decimal places.

WORKING

Answer _____

**6** A hopper has four equal trapezoid sides. Each has parallel sides of 1,500 mm and 300 mm, and a height of 900 mm. Find the total area of the four sides in m².

WORKING

Answer _____

**7** A duct 18 in by 12 in and 96 in long is open at both ends. Ignore seams. Find the area of sheet in ft².

WORKING

Answer _____

**8** A closed cylindrical tank is 1.2 m across and 2 m high. Find its surface area in m² to two decimal places.

WORKING

Answer _____

**9** A machine guard panel is 900 mm by 600 mm with a half-circle notch of radius 150 mm cut from one long edge. Find the area of the panel to two decimal places.

WORKING

Answer _____

**10** A sheet metal jacket wraps once around a pipe with an outside diameter of 168 mm. The jacket is 3,000 mm long. Ignore laps. Find the area of the jacket in m² to two decimal places.

WORKING

Answer _____

**11** A 300 mm by 200 mm access hole is cut in a 1,200 mm by 800 mm cover plate. Find the area of plate left in mm² and in m².

WORKING

Answer _____

**12** A shop floor is 14 ft by 9 ft 6 in. Find its area in ft².

WORKING

Answer _____

**EXERCISE 10C**

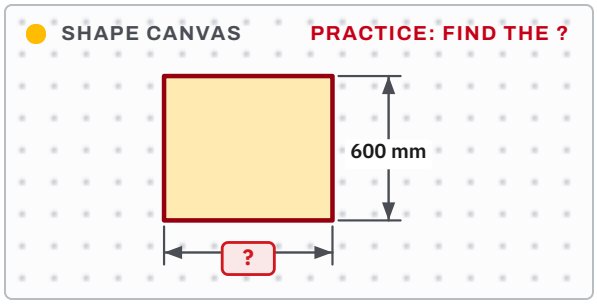
# Shape Canvas

Hidden sizes, break it down, unfold it, change one value

Name _____

Date _____

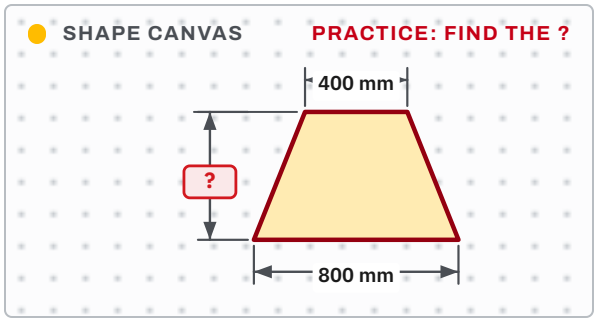
**1** The plate has an area of 0.42 m². Find the hidden length.




WORKING

Answer _____

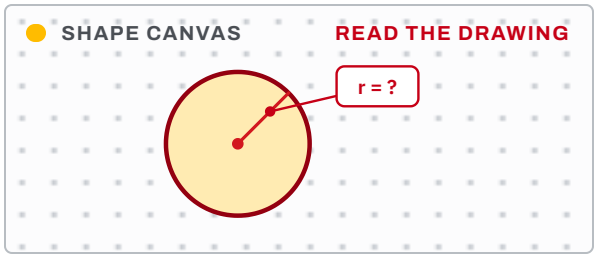
**2** The trapezoid has an area of 300,000 mm². Find the hidden height.




WORKING

Answer _____

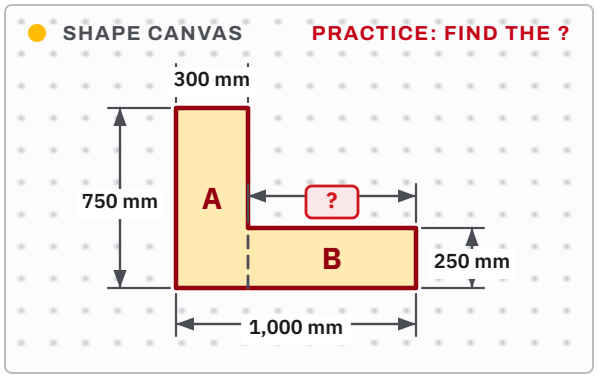
**3** The disc has an area of 31,416 mm². Find the hidden radius to the nearest mm.




WORKING

Answer _____

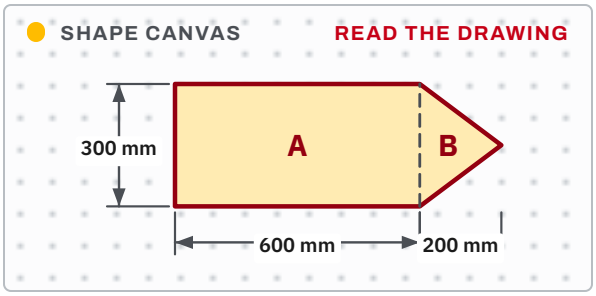
**4** Find the hidden length of the foot B, then the area of the L-plate.




WORKING

Answer _____

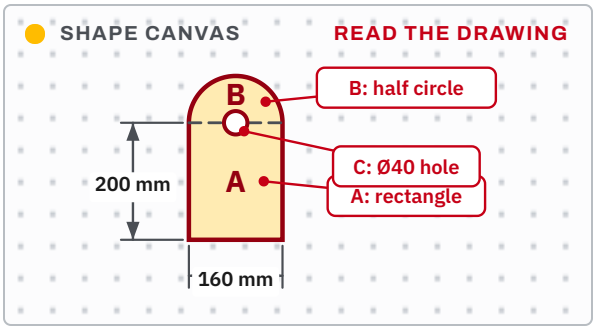
**5** Find the area of the plate with the pointed end.




WORKING

Answer _____

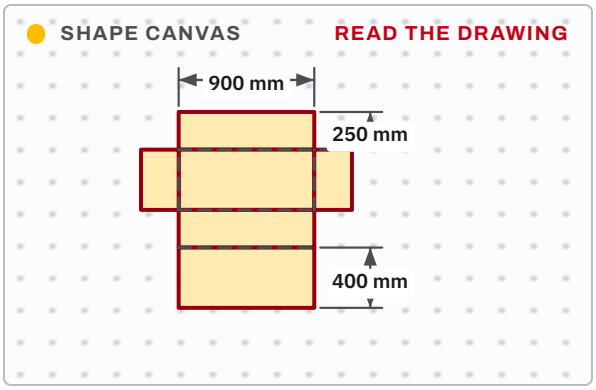
**6** Find the area of the lug, A + B − C, to two decimal places.




WORKING

Answer _____

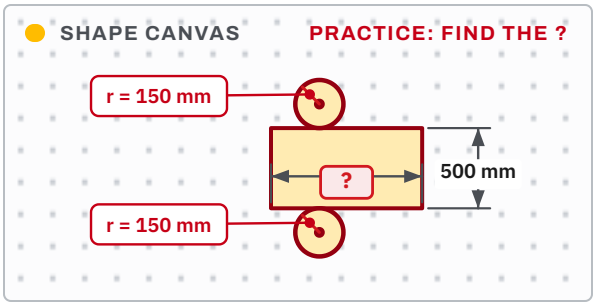
**7** Unfold it: this is the net of a closed box. Find the surface area of the box in mm² and m².




WORKING

Answer _____

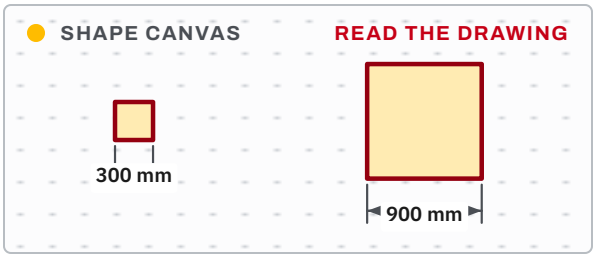
**8** This net makes a closed cylinder with a radius of 150 mm. Find the hidden stretch-out and the surface area, to two decimal places.




WORKING

Answer _____

**9** Change one value: the 300 mm side of a square plate is made 3 times longer. Find both areas. How many times bigger is the new area?




WORKING

Answer _____

EXERCISE 10D

# Materials takeoff

Openings, waste, whole units and cost

Name _____

Date _____

**1** A wall is 7.2 m by 2.4 m with a door 0.9 m by 2.1 m and a window 1.8 m by 1.2 m. Drywall sheets are 1.2 m by 2.4 m at $19.25. How many sheets, and what is the cost?

WORKING

Answer _____

**2** A ceiling is 4.8 m by 3.6 m. Drywall sheets are 1.2 m by 3.6 m. How many sheets?

WORKING

Answer _____

**3** A bathroom floor is 2.4 m by 1.8 m. The tiles are 300 mm by 600 mm. Allow 15% waste. How many tiles?

WORKING

Answer _____

**4** A backsplash is 3.0 m long and 0.6 m high. The tiles are 100 mm by 100 mm. Allow 10% waste. How many tiles?

WORKING

Answer _____

**5** A room is 5.0 m by 4.0 m with walls 2.5 m high. Leave out a door 0.9 m by 2.1 m and two windows 1.2 m by 1.2 m. Two coats. One can covers 35 m² and costs $49.99. How many cans, and what is the cost?

WORKING

Answer _____

**6** Two walls are each 30 ft long and 9 ft high. They get one coat. The label says 1 US gal covers 350 ft². How many whole gallons?

WORKING

Answer _____

**7** Laminate flooring for a room 15 ft by 12 ft. Allow 7% waste. One box covers 20 ft² and costs $44.90. How many boxes, and what is the cost?

WORKING

Answer _____

**8** A wall is 9.6 m by 2.7 m with two windows 1.5 m by 1.2 m. Siding comes in boxes that cover 2.32 m². Allow 10% waste. How many boxes?

WORKING

Answer _____

**9** Carpet is sold by the whole square metre at $32.50 a square metre. A room is 4.2 m by 3.5 m. How many square metres do you buy, and what is the cost?

WORKING

Answer _____

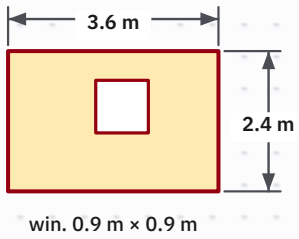
**10** You have 6 drywall sheets, each 2.88 m². A wall needs 13.5 m² plus 10% waste. Is that enough? By how much?

WORKING

Answer _____

**11** Drywall sheets are 1.2 m by 2.4 m. How many sheets do you buy for this wall?

SHAPE CANVAS — READ THE DRAWING




WORKING

Answer _____

EXERCISE 10K

# Carpentry and the BCBC

Sheathing, drywall, flooring and roofs, with BCBC 2024 Part 9 tables

Name _____

Date _____

**1** Studs are 600 mm o.c. Table 9.29.5.3. (Sentence 9.29.5.3.(1), page 9-275) lists gypsum board 9.5 mm, 12.7 mm and 15.9 mm thick. Which thicknesses may be used on these walls as a single layer?

WORKING

Answer _____

**2** Ceiling joists are 600 mm o.c. with a painted finish. Use Table 9.29.5.3. (page 9-275). May regular 12.7 mm gypsum board run parallel to the joists? Perpendicular?

WORKING

Answer _____

**3** Floor joists are 400 mm o.c. Table 9.23.15.5.-A (Sentence 9.23.15.5.(1), page 9-224) gives the minimum plywood subfloor thickness. What is it? How many 1.22 m by 2.44 m sheets cover a floor 9.6 m by 7.2 m, with no waste allowance?

WORKING

Answer _____

**4** The same floor is framed with joists at 600 mm o.c. What minimum plywood subfloor thickness does Table 9.23.15.5.-A (page 9-224) give, and how much thicker is it than at 400 mm o.c.?

WORKING

Answer _____

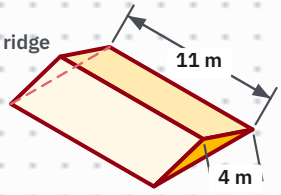
**5** Roof rafters are 600 mm o.c. on a roof that is not a walking deck. Table 9.23.16.7.-A (Sentence 9.23.16.7.(2), page 9-226) gives the minimum plywood roof sheathing thickness. What is it with the edges supported, and with the edges unsupported?

WORKING

Answer _____

**6** Sheathe both sides of the roof with 1.22 m by 2.44 m sheets. Allow 5% waste. How many sheets?

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

**7** Table 9.26.3.1. (Sentence 9.26.3.1.(1), page 9-242) sets a minimum slope for asphalt shingles of 1 in 3 for normal application and 1 in 6 for low slope application. A roof rises 4 in per 12 in. Write it as 1 in ?. Which application may be used?

WORKING

Answer _____

**8** Use the same table. A roof rises 3 in per 12 in. Write it as 1 in ?. May asphalt shingles be applied, and how?

WORKING

Answer _____

**9** A roof rises 2 in per 12 in. Write it as 1 in ?. What does Table 9.26.3.1. (page 9-242) allow for asphalt shingles?

WORKING

Answer _____

**10** A room is 14 ft by 12 ft with 8 ft walls. Drywall the walls and the ceiling with 4 ft by 8 ft sheets at $15.98. Leave out a door of 20 ft² and a window of 15 ft². Round the walls and the ceiling up separately. How many sheets, and what is the cost?

WORKING

Answer _____

**11** Table 9.23.17.2.-A (Sentence 9.23.17.2.(1), page 9-227) lists the minimum thickness of OSB, O-2 Grade, wall sheathing. What is it with studs 400 mm o.c., and with studs 600 mm o.c.?

WORKING

Answer _____

**12** Walls with a total length of 40 m are 2.4 m high. The openings add up to 9.5 m². Sheathing sheets are 1.22 m by 2.44 m. How many sheets?

WORKING

Answer _____

**13** A gable end wall is a rectangle 8 m wide and 2.4 m high with a triangle on top, 8 m at the base and 1.6 m high. Leave out a 1.2 m² window. Siding boxes cover 2 m². Allow 10% waste. How many boxes?

WORKING

Answer _____

# 11 Volume and Capacity

EXERCISES

Practice with cubic units, the volume of prisms, cylinders, cones, pyramids and spheres, composite solids, and capacity in litres and US gallons (1 US gal = 231 in³ = 3.785411784 L).

**EXERCISE 11A**

## Skills

One skill at a time

Name _____

Date _____

**1** A block of 1 cm cubes is 6 long, 4 wide and 3 high. Find its volume.

WORKING

Answer _____

**2** Write 3 cm³ in mm³.

WORKING

Answer _____

**3** Write 2,500,000,000 mm³ in m³.

WORKING

Answer _____

**4** Write 0.45 m³ in litres.

WORKING

Answer _____

**5** Write 3,456 in³ in ft³.

WORKING

Answer _____

**6** Write 2,000,000 mm³ in litres.

WORKING

Answer _____

**7** Box 700 mm by 400 mm by 250 mm. Find the volume in m³.

WORKING

Answer _____

**8** Cube with 0.6 m edges. Find the volume in m³.

WORKING

Answer _____

**9** Triangular prism: end base 300 mm, end height 200 mm, length 1,500 mm. Find the volume in litres.

WORKING

Answer _____

**10** Write 1,155 in³ in US gallons.

WORKING

Answer _____

**11** Write 75 US gal in litres, to two decimal places.

WORKING

Answer _____

**12** Write 500 L in US gallons, to two decimal places.

WORKING

Answer _____

**13** Cylinder: radius 150 mm, height 600 mm. Capacity in litres, to two decimal places?

WORKING

Answer _____

**14** Cone: radius 200 mm, height 450 mm. Capacity in litres, to two decimal places?

WORKING

Answer _____

**15** Square pyramid: base 900 mm by 900 mm, height 600 mm. Capacity in litres?

WORKING

Answer _____

**16** Sphere: radius 250 mm. Capacity in litres, to two decimal places?

WORKING

Answer _____

**17** Hemisphere: radius 0.4 m. Volume in $m^3$, to three decimal places?

WORKING

Answer _____

**18** Pipe with an inside diameter of 80 mm. Litres per metre, to two decimal places?

WORKING

Answer _____

**19** A box holds 0.24 $m^3$. Its base is 0.8 m by 0.5 m. Find its height.

WORKING

Answer _____

**20** How many US gallons are in 1 $ft^3$ (1,728 $in^3$), to two decimal places?

WORKING

Answer _____

**21** Which holds more, a 40 L drum or a 10 US gal can? Show the litres to two decimal places.

WORKING

Answer _____

**22** Cylinder: radius 0.2 m, height 0.5 m. Volume in $m^3$, to three decimal places?

WORKING

Answer _____

EXERCISE 11B

# Shop problems

Tanks, pipes and parts, metric and imperial

Name _____

Date _____

**1** A hydraulic reservoir is 800 mm by 500 mm by 450 mm inside. How many litres does it hold? How many 20 L pails of oil fill it?

WORKING

Answer _____

**2** A water heater tank is a cylinder 18 in across and 48 in tall inside. Find its capacity in US gallons, to two decimal places.

WORKING

Answer _____

**3** Before a pressure test, 75 m of pipe with an inside diameter of 52 mm is filled with water. How many litres, to two decimal places?

WORKING

Answer _____

**4** A system has 40 m of pipe with a 100 mm inside diameter and 25 m with a 50 mm inside diameter. Find the total litres, to two decimal places.

WORKING

Answer _____

**5** A sump is 1.5 m by 1.0 m. The water in it is 0.6 m deep. A pump removes 45 L per minute. How many litres are in the sump, and how long does pumping take?

WORKING

Answer _____

**6** A hopper is a cylinder 1.2 m across and 1.0 m high on a cone 0.8 m high. Find its capacity in litres, to two decimal places.

WORKING

Answer _____

**7** A drum is 22.5 in across and 33.5 in tall inside. Find its capacity in US gallons, to two decimal places.

WORKING

Answer _____

**8** A tank is a cylinder with a hemisphere on each end. The cylinder part is 2 m long. The radius is 0.5 m. Find its capacity in litres, to two decimal places.

WORKING

Answer _____

**9** A vertical tank is 1 m across inside and holds water 1.4 m deep. How many litres are in it, to two decimal places?

WORKING

Answer _____

**10** A solid round bar is 50 mm in diameter and 3 m long. Find its volume in $cm^3$, to two decimal places.

WORKING

Answer _____

**11** A tube has an outside diameter of 114 mm, an inside diameter of 102 mm and a length of 6,000 mm. Find the volume of metal in $cm^3$, to two decimal places.

WORKING

Answer _____

**12** A fuel day tank is 36 in by 24 in by 20 in inside. Find its capacity in US gallons to two decimal places. A heater burns 2.5 US gal per hour. How many hours will a full tank last, to the nearest hour?

WORKING

Answer _____

**13** A blast-media hopper is an upside-down square pyramid. The top is 1,200 mm square and the depth is 900 mm. Find its capacity in litres.

WORKING

Answer _____

EXERCISE 11C

# Shape Canvas

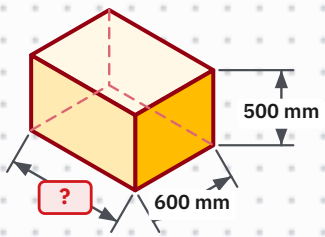
Hidden sizes, break it down, unfold it, change one value

Name _____

Date _____

**1** The tank holds 240 L. Find the hidden length.

SHAPE CANVAS PRACTICE: FIND THE ?

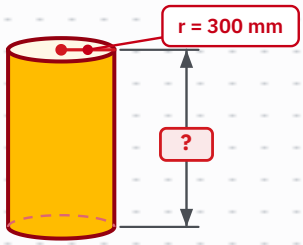



WORKING

Answer _____

**2** The cylinder holds 282.74 L to two decimal places. Find the hidden height to the nearest mm.

SHAPE CANVAS PRACTICE: FIND THE ?

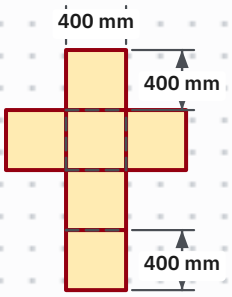



WORKING

Answer _____

**3** Fold the net into a cube. Find its capacity in litres.

SHAPE CANVAS READ THE DRAWING

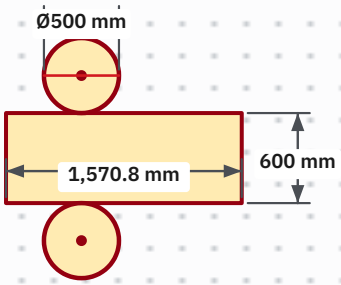



WORKING

Answer _____

**4** Fold the net into a closed cylinder. Find its capacity in litres, to two decimal places.

SHAPE CANVAS READ THE DRAWING

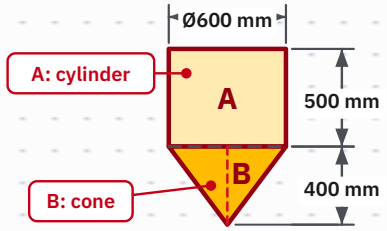



WORKING

Answer _____

**5** Break it down: A is a cylinder 600 mm across and B is a cone. Find the capacity of the hopper, A + B, in litres to two decimal places.

SHAPE CANVAS READ THE DRAWING

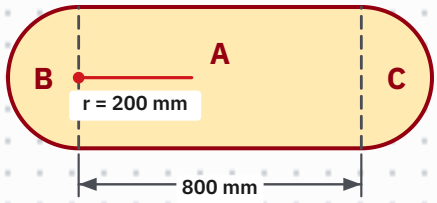



WORKING

Answer _____

**6** The tank has a hemisphere on each end. Find its capacity in litres, to two decimal places.

SHAPE CANVAS READ THE DRAWING

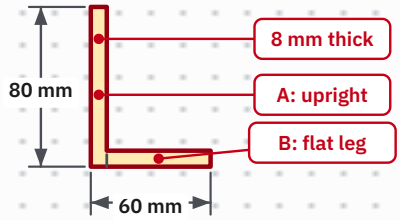



WORKING

Answer _____

**7** Break down the L-shaped end of the angle bar into two rectangles. Find the end area and the volume in cm³.

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

**8** Change one value: a spherical float with a radius of 100 mm has its radius doubled. Find both capacities in litres, to two decimal places. How many times bigger is the new one?

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

**9** Change one value: a cone has a radius of 300 mm and a height of 300 mm. The height is doubled. Find both capacities in litres, to two decimal places.

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

EXERCISE 11D

# Concrete, steel and lumber

Cubic metres and yards, mass, board feet and tank capacity

Name _____

Date _____

**1** A patio slab is 4.8 m by 3.6 m and 100 mm thick. Concrete in m³ and in cubic yards, to two decimal places?

WORKING

Answer _____

**2** A driveway is 30 ft by 12 ft and 4 in thick. How many cubic yards, to two decimal places?

WORKING

Answer _____

**3** A strip footing is 450 mm wide, 150 mm thick and 32 m long. Concrete in m³?

WORKING

Answer _____

**4** Ten round piers are each 200 mm across and 1.0 m deep. Concrete in m³, to two decimal places?

WORKING

Answer _____

**5** Find the mass of the 2.16 m³ footing from question 3 at 2,240 kg/m³ and at 2,400 kg/m³, the range FHWA gives for normal concrete.

WORKING

Answer _____

**6** A steel plate is 1,500 mm by 600 mm and 12 mm thick. Mass at 7,850 kg/m³, to two decimal places?

WORKING

Answer _____

**7** A steel round bar is 40 mm across and 3 m long. Mass at 7,850 kg/m³, to two decimal places?

WORKING

Answer _____

**8** An order is 16 pieces of 2 × 8, each 14 ft long, at $1,250 per MBF. Board feet and cost, to two decimal places?

WORKING

Answer _____

**9** A 2 × 4 is 10 ft long. Find its board feet from the nominal size. Then find the board feet of wood in the dressed piece, 1½ in by 3½ in (Canadian Wood Council S-Dry size). Two decimal places.

WORKING

Answer _____

**10** A tank is 2,000 mm by 1,200 mm by 1,000 mm inside. Capacity in litres, US gallons and imperial gallons, to two decimal places?

WORKING

Answer _____

**11** A vertical tank is 1,500 mm across and 2,000 mm high inside. Capacity in litres, US gallons and imperial gallons, to two decimal places?

WORKING

Answer _____

**12** An old tank is stamped 500 imperial gal. How many litres, and how many US gallons, to two decimal places?

WORKING

Answer _____

EXERCISE 11K

# Carpentry and the BCBC

Footings, foundation walls, slabs, granular base and backfill, with BCBC 2024 Sections 9.15 and 9.16

Name _____

Date _____

**1** Table 9.15.3.4. (Sentence 9.15.3.4.(1), page 9-120) gives minimum strip footing widths. What are they under exterior walls for 1, 2 and 3 floors supported?

WORKING

Answer _____

**2** Use the same table for interior walls, 1, 2 and 3 floors supported. For how many floors are the exterior and interior widths the same?

WORKING

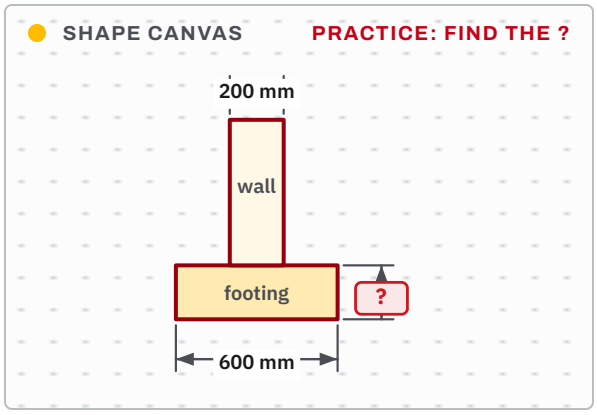
Answer _____

**3** A two-storey house has exterior strip footings at the minimum width, centred under 200 mm walls. Find the projection, then the minimum thickness from Sentence 9.15.3.8.(1) (page 9-121).

WORKING

Answer _____

**4** The drawings show a 600 mm footing centred under a 200 mm wall. Use Sentence 9.15.3.8. (1) to find the minimum thickness.




WORKING

Answer _____

**5** The footing from question 3 (350 mm by 100 mm) runs 44 m. Concrete in m³ and cubic yards, to two decimal places?

WORKING

Answer _____

**6** Table 9.15.3.4. gives 0.75 m² as the minimum column footing area for 2 floors supported (columns 3 m o.c.). A square pad must have at least that area. What is its smallest side, rounded up to the next whole mm?

WORKING

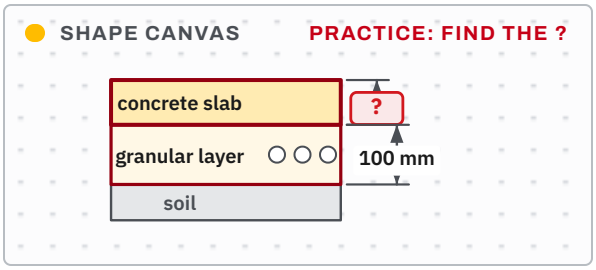
Answer _____

**7** Four square pads are 867 mm by 867 mm and 200 mm thick. Concrete in m³, to two decimal places?

WORKING

Answer _____

**8** A basement slab is 10.8 m by 8.4 m at the minimum thickness of Sentence 9.16.4.3.(1) (page 9-130). Find the thickness, then the concrete in m³ and cubic yards, to two decimal places.




WORKING

Answer _____

**9** Under the same slab, Sentence 9.16.2.1.(1) (page 9-129) sets the least depth of coarse clean granular material. What is it, and how many m³ are needed?

WORKING

Answer _____

**10** Must granular material be placed under the slab of a detached garage? Use Sentence 9.16.2.1.(2) (page 9-129).

WORKING

Answer _____

**11** The top of a footing is 1.8 m below finished ground. Sentence 9.15.4.6.(1) (page 9-127) says exterior foundation walls extend not less than 150 mm above finished ground. What is the least wall height? Find the concrete for a 200 mm wall of that height, 36 m long.

WORKING

Answer _____

**12** A trench 36 m long, 0.8 m wide and 2.0 m deep holds a 450 mm by 150 mm footing and a 200 mm wall that rises from the footing past the top of the trench. How much backfill fills the trench?

WORKING

Answer _____

# 12 Angles and Right Triangles

EXERCISES

Practice with degrees, angle names, missing angles in triangles, the Pythagorean theorem, the 3-4-5 check and diagonals.

**EXERCISE 12A**

## Skills

Angles, triangles and the Pythagorean theorem, one skill at a time.

Name _____

Date _____

**1** How many degrees are in one third of a turn?

WORKING

Answer _____

**2** Nine holes are evenly spaced around a circle. What is the angle between neighbours?

WORKING

Answer _____

**3** Name each angle: 12°, 90°, 179°, 95°.

WORKING

Answer _____

**4** Two angles sit on a straight line. One is 64°. Find the other.

WORKING

Answer _____

**5** Four angles go around a point: 80°, 95°, 110° and ?. Find the ?

WORKING

Answer _____

**6** A triangle has angles of 52° and 71°. Find the third angle.

WORKING

Answer _____

**7** One acute angle of a right triangle is 38°. Find the other.

WORKING

Answer _____

**8** A triangle has two equal sides with 64° between them. Find the other two angles.

WORKING

Answer _____

**9** A triangle has two equal angles of 35°. Find the third angle.

WORKING

Answer _____

**10** Legs of 700 mm and 2,400 mm. Find the hypotenuse.

WORKING

Answer _____

**11** Legs of 350 mm and 480 mm. Find the hypotenuse to two decimal places.

WORKING

Answer _____

**12** Hypotenuse 1,700 mm, one leg 800 mm. Find the other leg.

WORKING

Answer _____

**13** Hypotenuse 2.6 m, one leg 1.4 m. Find the other leg to two decimal places.

WORKING

Answer _____

**14** Legs of 14″ and 19″. Find the hypotenuse to the nearest 1/16″.

WORKING

Answer _____

**15** Do 8, 15 and 17 make a right triangle? Show the check.

WORKING

Answer _____

**16** Do 6, 9 and 11 make a right triangle? Show the check.

WORKING

Answer _____

**17** A rectangle is 1,100 mm by 600 mm. Find the diagonal to the nearest millimetre.

WORKING

Answer _____

**18** A square has 450 mm sides. Find the diagonal to two decimal places.

WORKING

Answer _____

EXERCISE 12B

## Shop problems

Layout, bracing and squaring in the shop. Metric and imperial.

Name _____

Date _____

**1** A pump flange has 12 bolt holes evenly spaced on its bolt circle. What is the angle between neighbouring holes? What is the angle at the centre from hole 1 to hole 4?

WORKING

Answer _____

**2** A flat bar is bent so that one side of the bend makes 142° with the straight part. What angle is on the other side of the bend? Is it acute or obtuse?

WORKING

Answer _____

**3** A corner brace on a machine stand runs from a point 450 mm up the leg to a point 600 mm along the rail. Both points are measured from the inside corner. How long is the brace, centre line to centre line? Two braces are cut from one bar with two cuts, 3 mm kerf each. How much bar do they use?

WORKING

Answer _____

**4** You square up a skid on the floor. You mark 1.2 m along one side and 1.6 m along the other. The tape reads 1.998 m between the marks. What should it read? Is the corner open or closed?

WORKING

Answer _____

**5** A pipe rack frame is 3,000 mm by 1,800 mm. The diagonals measure 3,505 mm and 3,492 mm. What should each diagonal measure, to the nearest millimetre? Which two corners do you push toward each other?

WORKING

Answer _____

**6** A sloped support covers 2,250 mm along the floor and rises 800 mm. How long is it, to two decimal places?

WORKING

Answer _____

**7** A gate frame is 4′ 0″ wide and 6′ 6″ tall. Find the length of the diagonal brace, corner to corner, to the nearest 1/16″. Give it in feet and inches.

WORKING

Answer _____

**8** A guard is bent from one bar into a right triangle. The two short sides are 18″ and 24″. How long is the bar before bending? Ignore the bends.

WORKING

Answer _____

**9** A three-arm bracket has its arms evenly spaced around a hub. What is the angle between arms? If one arm is moved so one of the angles next to it is 100°, what are the other two angles?

WORKING

Answer _____

**EXERCISE 12C**

## Shape Canvas

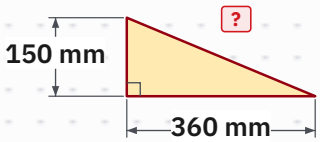
Find every red ?. Break it down where the drawing shows a dashed line.

Name _____

Date _____

**1** Find the hypotenuse.

SHAPE CANVAS PRACTICE: FIND THE ?

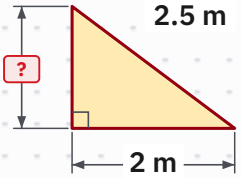



WORKING

Answer _____

**2** Find the missing leg.

SHAPE CANVAS PRACTICE: FIND THE ?

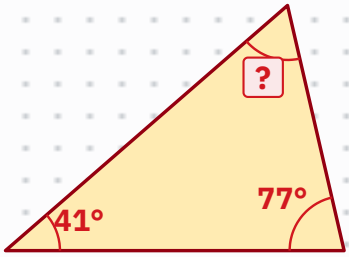



WORKING

Answer _____

**3** Find the third angle.

SHAPE CANVAS PRACTICE: FIND THE ?

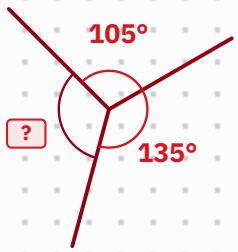



WORKING

Answer _____

**4** Find the ?

SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

**5** Find the ? on the bent strap.

SHAPE CANVAS PRACTICE: FIND THE ?

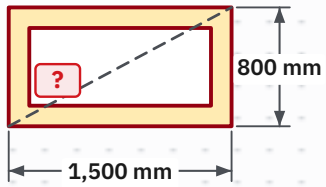



WORKING

Answer _____

**6** Find the diagonal of the frame.

SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

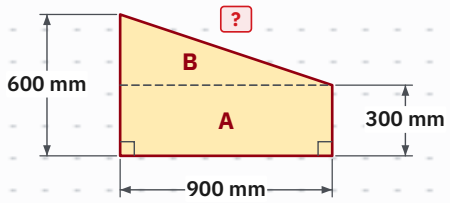
**7** Change one value. A 300-400-500 triangle has its 300 mm leg made 600 mm long. The other leg stays 400 mm. Find the new hypotenuse to two decimal places. Did it double?

WORKING

Answer _____

**8** Break it down. The dashed line splits the plate into rectangle A and right triangle B. Find the legs of triangle B, then the sloped edge to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

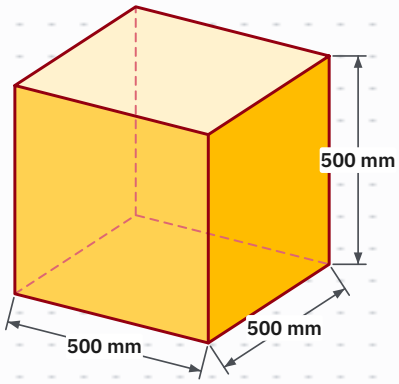



WORKING

Answer _____

**9** Break it down. A cube has 500 mm edges. Find the diagonal of one face. Then use it and an edge as legs to find the longest rod that fits inside, corner to corner. Two decimal places.

SHAPE CANVAS READ THE DRAWING




WORKING

Answer _____

**EXERCISE 12D**

# Angles and DMS

Angle pairs, parallel lines, and degrees, minutes and seconds.

Name _____

Date _____

**1** Find the complement of 34°.

WORKING

Answer _____

**2** Find the supplement of 117°.

WORKING

Answer _____

**3** Two lines cross. Find each ?

SHAPE CANVAS — PRACTICE: FIND THE ?

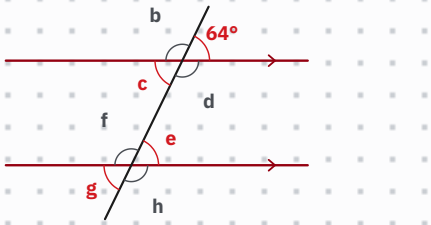



WORKING

Answer _____

**4** The two level lines are parallel. Angle a is 64°. Find angles d, e and f.

SHAPE CANVAS — READ THE DRAWING




WORKING

Answer _____

**5** A brace crosses two parallel rails. One alternate angle between the rails is 58°. What is the other?

WORKING

Answer _____

**6** A brace crosses two parallel rails. One co-interior angle is 109°. What is the other?

WORKING

Answer _____

**7** Change 52°18′ to decimal degrees.

WORKING

Answer _____

**8** Change 19°44′24″ to decimal degrees.

WORKING

Answer _____

**9** Change 28.65° to degrees, minutes and seconds.

WORKING

Answer _____

**10** Change 71.2525° to degrees, minutes and seconds.

WORKING

Answer _____

**11** Add 33°27′48″ + 41°50′35″.

WORKING

Answer _____

**12** Find 180° − 96°42′15″.

WORKING

Answer _____

**EXERCISE 12E**

# Stairs

Risers, treads, total rise and run, and the stringer line.

Name _____

Date _____

**1** A stair has 15 risers of 188 mm and 14 treads with a 270 mm unit run. Find the total rise, the total run and the stringer line to two decimal places.

WORKING

Answer _____

**2** A private stair has a total rise of 2,560 mm. Use the 200 mm maximum rise to find the number of risers, then the unit rise to two decimal places.

WORKING

Answer _____

**3** A public stair has a total rise of 2,880 mm. Using the 180 mm maximum rise and the 280 mm minimum run, find the number of risers, the number of treads, the total run and the stringer line to two decimal places.

WORKING

Answer _____

**4** A stair has 8 risers of $7\frac{3}{8}$″ and 7 treads of $10\frac{1}{4}$″. Find the total rise, the total run and the stringer line to the nearest 1/16″.

WORKING

Answer _____

**5** The risers of one flight measure 182, 185, 183, 187, 190 and 186 mm. Do they meet Article 9.8.4.4, Sentence (1): 5 mm between adjacent risers and 10 mm between the tallest and shortest?

WORKING

Answer _____

**6** The tread runs of one flight measure 265, 268, 262, 271 and 266 mm. Do they meet Article 9.8.4.4, Sentence (3)?

WORKING

Answer _____

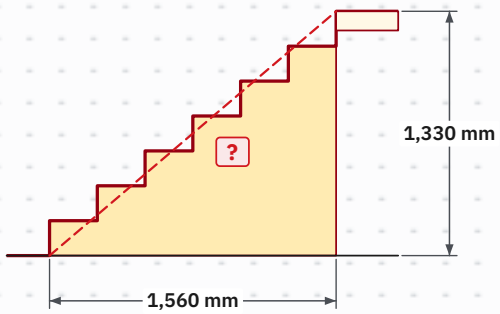
**7** A rectangular tread has a run of 255 mm and a depth of 290 mm. Does it meet Sentence 9.8.4.2.(2)? What is the largest depth allowed?

WORKING

Answer _____

**8** Find the stringer line, bottom corner to top corner, to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

**EXERCISE 12K**

## Carpentry and the BCBC

Rafters, layout and stairs. Code values are from the BC Building Code 2024, Division B.

Name _____

Date _____

**1** A building is 8,400 mm wide with the ridge in the middle and a 6 in 12 roof. Find the run, the rise and the rafter line length to two decimal places.

WORKING

Answer _____

**2** A 4 in 12 roof has a rafter run of 13′ 0″. Find the rise and the rafter line length to the nearest 1/16″.

WORKING

Answer _____

**3** Find the length per foot of run of a 9 in 12 roof. Use it to find the rafter line length for a 10′ run, in feet and inches.

WORKING

Answer _____

**4** A deck frame is 4.2 m by 3.0 m. What should each diagonal measure, to the nearest millimetre?

WORKING

Answer _____

**5** You square a wall layout with 3-4-5 using 900 mm as the unit. Where are the two marks, and how far apart should they be?

WORKING

Answer _____

**6** A wall layout is 7,800 mm by 3,000 mm. The diagonals measure 8,362 mm and 8,349 mm. What should they be, to the nearest millimetre? Which corners do you move?

WORKING

Answer _____

**7** A private stair has a total rise of 2,450 mm. Using the 200 mm maximum rise in Table 9.8.4.1 (Article 9.8.4.1, page 9-31), find the number of risers and the unit rise to two decimal places. Is the unit rise above the 125 mm minimum?

WORKING

Answer _____

**8** A public stair is drawn with a unit rise of 185 mm. Does it meet Table 9.8.4.1 (page 9-31)? For a total rise of 2,700 mm, how many risers does it need at the public stair maximum?

WORKING

Answer _____

**9** Measured straight up from the nosing line, a beam is 1,980 mm above a stair. Article 9.8.2.2 (page 9-30) asks for 2,050 mm, or 1,950 mm over stairs serving a single dwelling unit or a house with a secondary suite. Is the clear height enough for a stair in a house? For a public stair?

WORKING

Answer _____

**10** The risers of a flight measure 192, 196, 193, 199 and 195 mm. Do they meet Article 9.8.4.4, Sentence (1) (page 9-32)?

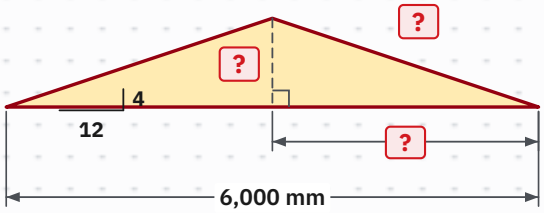
WORKING

Answer _____

**11** A building is 6,000 mm wide with the ridge in the middle and a 4 in 12 roof. Find each ? to two decimal places.

SHAPE CANVAS

PRACTICE: FIND THE ?




WORKING

Answer _____

# 13 Trigonometry

EXERCISES

Practice naming sides, using sine, cosine and tangent, finding sides and angles, and solving slopes, pipe offsets, bevels and mitres. Your calculator must be in DEG mode.

**EXERCISE 13A**

## Skills

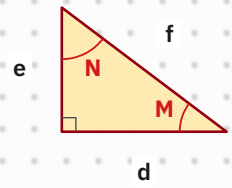
Sides, ratios, missing sides and missing angles.

Name _____

Date _____

**1** Look at the triangle. Which side is opposite angle M? Which side is adjacent to angle M? Which side is the hypotenuse? Which side is opposite angle N?

SHAPE CANVAS — READ THE DRAWING




WORKING

Answer _____

**2** Find sin 25°, cos 25° and tan 25° to four decimal places.

WORKING

Answer _____

**3** A right triangle has sides 8, 15 and 17. Angle A is opposite the 8. Write sin A, cos A and tan A to four decimal places.

WORKING

Answer _____

**4** Hypotenuse 650 mm, angle 32°. Find the opposite side to two decimal places.

WORKING

Answer _____

**5** Hypotenuse 1,250 mm, angle 48°. Find the adjacent side to two decimal places.

WORKING

Answer _____

**6** Adjacent 900 mm, angle 24°. Find the opposite side to two decimal places.

WORKING

Answer _____

**7** Opposite 520 mm, angle 38°. Find the hypotenuse to two decimal places.

WORKING

Answer _____

**8** Opposite 300 mm, angle 61°. Find the adjacent side to two decimal places.

WORKING

Answer _____

**9** Hypotenuse 42″, angle 15°. Find the opposite side to the nearest 1/16″.

WORKING

Answer _____

**10** Opposite 350 mm, adjacent 800 mm. Find the angle to two decimal places.

WORKING

Answer _____

**11** Opposite 0.45 m, hypotenuse 1.2 m. Find the angle to two decimal places.

WORKING

Answer _____

**12** Adjacent 700 mm, hypotenuse 950 mm. Find the angle to two decimal places.

WORKING

Answer _____

**13** One acute angle of a right triangle is 27.4°. Find the other one.

WORKING

Answer _____

**14** A slope rises 120 mm over a run of 6,000 mm. Find the percent grade and the angle to two decimal places.

WORKING

Answer _____

**15** tan A = 0.25. Find A to two decimal places.

WORKING

Answer _____

EXERCISE 13B

# Shop problems

Pipe offsets, slopes, bevels and mitres. Metric and imperial.

Name _____

Date _____

**1** A line moves over 375 mm with 45° fittings. Find the travel and the run to two decimal places.

WORKING

Answer _____

**2** A line moves over 9″ with 60° fittings. Find the travel and the run to the nearest 1/16″.

WORKING

Answer _____

**3** A line moves over 12″ with 22.5° fittings. Find the travel to the nearest 1/16″.

WORKING

Answer _____

**4** An offset made with 30° fittings has a travel of 640 mm. Find the set and the run to two decimal places.

WORKING

Answer _____

**5** A line must move over 250 mm in a run of 433 mm. What fitting angle does it need, to two decimal places?

WORKING

Answer _____

**6** A rolling offset rises 450 mm and rolls 600 mm. Find the true offset, then the travel with 45° fittings, to two decimal places.

WORKING

Answer _____

**7** A ramp must rise 750 mm. The job sheet sets its angle at 4.8°. How long is the run, to the nearest millimetre?

WORKING

Answer _____

**8** A drain line is laid at a 1% grade over an 18 m run. How far does it fall, in millimetres?

WORKING

Answer _____

**9** A conveyor 8,500 mm long slopes up at 12°. Find the rise and the floor length to two decimal places.

WORKING

Answer _____

**10** A 25 mm plate is bevelled at 30° through its thickness. How wide is the bevel at the top, to two decimal places? What groove angle do two such plates open?

WORKING

Answer _____

**11** A frame has 10 equal sides. Find the corner angle, the cut angle to the edge and the angle away from a square cut.

WORKING

Answer _____

**12** A truss top chord rises at 18° over a half span of 3,600 mm. Find the rise and the chord length to two decimal places.

WORKING

Answer _____

**13** A 14 mm plate has a bevel 8 mm wide at the top, through the full thickness. Find the bevel angle to two decimal places.

WORKING

Answer _____

**14** With the 45° travel constant 1.414, what travel do you get for a 375 mm set? How far is it from the calculator value in problem 1, to two decimal places?

WORKING

Answer _____

**EXERCISE 13C**

# Shape Canvas

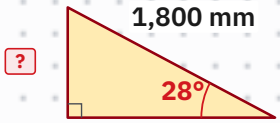
Every triangle is drawn at its true angle. Find every red ?

Name _____

Date _____

**1** Find the height to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

**2** Find the hypotenuse to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

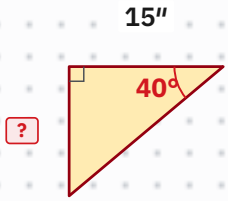



WORKING

Answer _____

**3** Find the opposite side to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?

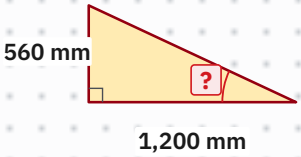



WORKING

Answer _____

**4** Find the angle to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

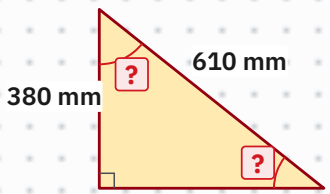



WORKING

Answer _____

**5** Find both acute angles to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

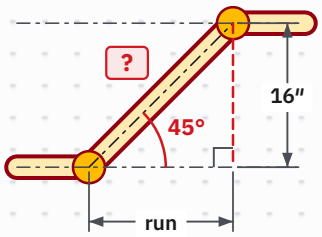



WORKING

Answer _____

**6** A 45° offset. Find the travel to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?

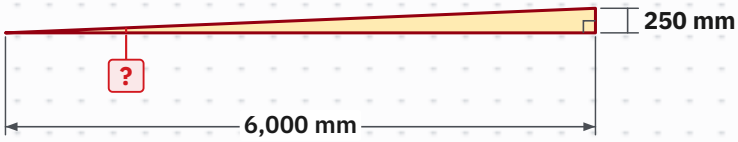



WORKING

Answer _____

**7** A ramp rises 250 mm over a run of 6,000 mm. Find its angle to two decimal places. The drawing is to scale.

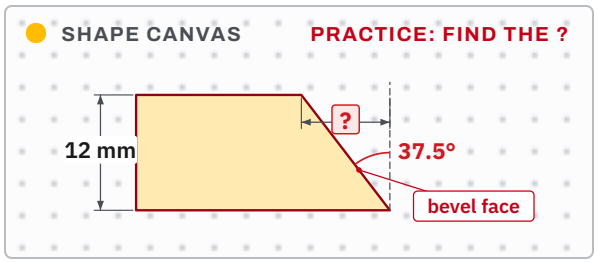
SHAPE CANVAS PRACTICE: FIND THE ?




WORKING

Answer _____

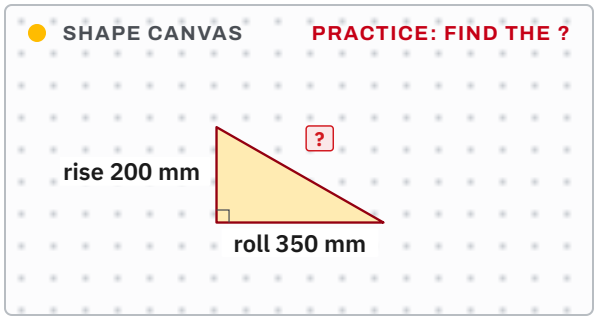
**8** Find the bevel width to two decimal places.




WORKING

Answer _____

**9** Break it down. A rolling offset rises 200 mm and rolls 350 mm. Find the true offset, then the travel with 60° fittings. Two decimal places.




WORKING

Answer _____

**10** Change one value. A 45° offset with a 300 mm set has a travel of 424.26 mm. Double the set to 600 mm. What is the travel, to two decimal places? Did it double?

WORKING

Answer _____

**EXERCISE 13K**

# Carpentry and the BCBC

Roof slopes, inches per foot, stairs, ramps and rafter cuts. Code values are from the BC Building Code 2024, Division B.

Name _____

Date _____

**1** Find the angle of a 3 in 12 roof and of a 9 in 12 roof, to two decimal places.

WORKING

Answer _____

**2** A drain slopes $\frac{1}{2}$″ per foot. Write it as 1 in …, as a percent grade and as an angle, to two decimal places.

WORKING

Answer _____

**3** A roof rises $7\frac{1}{2}$″ per foot of run. Write it as rise in 12 and as 1 in …, then find its angle to two decimal places.

WORKING

Answer _____

**4** A stair has a unit rise of $7\frac{1}{4}$″ and a unit run of $10\frac{1}{2}$″. Find the stair angle to two decimal places.

WORKING

Answer _____

**5** A roof is 3.5 in 12. Table 9.26.3.1 (Sentence 9.26.3.1.(1), page 9-242) gives a minimum slope for asphalt shingles of 1 in 3 for normal application and 1 in 6 for low slope application, and 1 in 4 for wood shingles. Which of the three may go on this roof?

WORKING

Answer _____

**6** A 2 in 12 roof framed with trusses covers an insulated ceiling of 150 m². Sentence 9.19.1.2.(1) (page 9-136) asks for a vent area of at least 1/300 of the insulated ceiling area. Sentence (2) asks for 1/150 where the roof slope is less than 1 in 6. Which Sentence applies, and what is the least vent area?

WORKING

Answer _____

**7** An interior ramp in a residential occupancy rises 380 mm at the steepest slope Clause 9.8.5.4.(1)(b) allows, 1 in 10 (page 9-33). Find the run, the length along the slope to two decimal places and the angle.

WORKING

Answer _____

**8** A ramp in an occupancy covered by Clause 9.8.5.4.(1)(d), 1 in 8 (page 9-34), has 4,000 mm of run. What is the most it can rise? What is its angle?

WORKING

Answer _____

**9** A rafter is 5 in 12. Find the roof angle, the angle between the seat cut and the rafter edge, and the angle between the plumb cut and the rafter edge.

WORKING

Answer _____

**10** Find the steepest stair angle that the public stair limits allow: the 180 mm maximum rise (Table 9.8.4.1) with the 280 mm minimum run (Table 9.8.4.2), page 9-31.

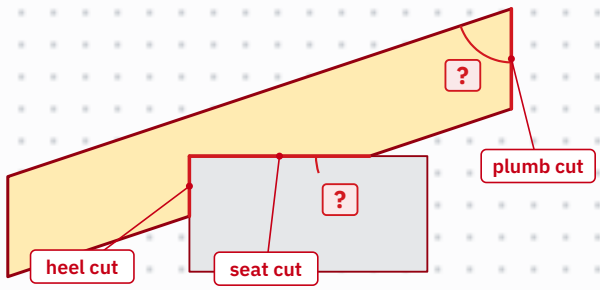
WORKING

Answer _____

**11** The rafter is 4 in 12. Find each ? to two decimal places.

SHAPE CANVAS

PRACTICE: FIND THE ?




WORKING

Answer _____

# A Kitchen Math

EXERCISES

Scaling recipes, cooking measures and food yield percent. Use a 250 mL cup, a 15 mL tablespoon and a 5 mL teaspoon.

**EXERCISE A1**

## Kitchen math

Scaling factors, cups and spoons, pounds and ounces, and yield percent

Name _____

Date _____

**1** A recipe makes 15 portions. Find the scaling factor to make 54 portions.

WORKING

Answer _____

**2** A recipe makes 60 portions. Find the scaling factor to make 45 portions.

WORKING

Answer _____

**3** A curry for 18 portions uses 2.7 kg of chicken. How much chicken for 42 portions?

WORKING

Answer _____

**4** A recipe makes 25 portions of 160 g. Find the scaling factor to make 70 portions of 200 g, and the new amount of a 1.25 kg ingredient.

WORKING

Answer _____

**5** Change $3\frac{1}{2}$ cups to millilitres, using a 250 mL cup.

WORKING

Answer _____

**6** A glaze needs 5 Tbsp and 1 tsp of honey. Give the amount in millilitres and in teaspoons.

WORKING

Answer _____

**7** Change 7 lb of carrots to kilograms. Round to three decimal places.

WORKING

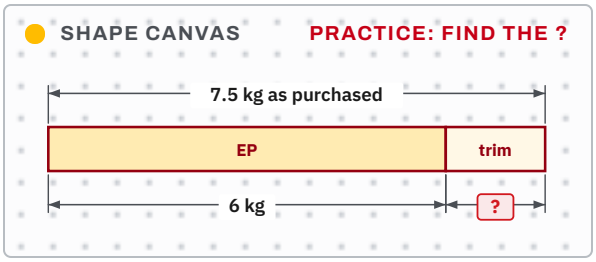
Answer _____

**8** Change 20 oz of cheese to grams. Round to the nearest gram.

WORKING

Answer _____

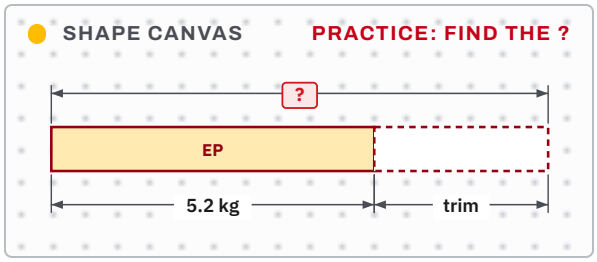
**9** A yield test on a 7.5 kg case of potatoes. Find the trim and the yield percent.




WORKING

Answer _____

**10** The yield percent is 65%. Find the amount as purchased.




WORKING

Answer _____

**11** A kitchen's yield test on pineapple gave 52%. How much pineapple is ready to serve from 15 kg as purchased?

WORKING

Answer _____

**12** A recipe needs 6 kg of trimmed leeks. The yield is 58%. How many kilograms must be bought? Round up to the next whole kilogram.

WORKING

Answer _____

**13** Salmon costs $18.00 per kilogram as purchased. The yield after filleting is 60%. Find the cost per kilogram of EP.

WORKING

Answer _____

**14** Change one value: in Problem 13 the yield improves to 72%. Find the new cost per kilogram of EP. Is it more or less?

WORKING

Answer _____

# B Practice Assessment

EXERCISES

Forty problems in the same skill order as the ACCESS math assessment: whole numbers, rounding, fractions, decimals, conversions and percent of a number. Work every problem by hand.

**EXERCISE B1**

## Practice assessment

No calculator. Whole numbers, rounding, fractions, decimals, conversions and percent

Name _____

Date _____

**NO CALCULATOR** Show your work in each box. Write a remainder with R, for example 12 R 3. Give fractions in lowest terms, and as mixed numbers when they are more than 1.

### Whole numbers

**1** Add: 3,486 + 927 + 12,058 + 674

WORKING

Answer _____

**2** Add: 40,503 + 8,769 + 315 + 2,097

WORKING

Answer _____

**3** Subtract: 7,004 − 2,867

WORKING

Answer _____

**4** Subtract: 30,050 − 14,786

WORKING

Answer _____

**5** Multiply: 486 × 307

WORKING

Answer _____

**6** Multiply: 729 × 584

WORKING

Answer _____

**7** Divide: 6,437 ÷ 8

WORKING

Answer _____

**8** Divide: 9,215 ÷ 23

WORKING

Answer _____

**9** Divide: 15,628 ÷ 47

WORKING

Answer _____

## Rounding

**10** Round 18.4762 to the nearest hundredth.

WORKING

Answer _____

**11** Round 0.8349 to the nearest hundredth.

WORKING

Answer _____

**12** Round 7.30951 to the nearest thousandth.

WORKING

Answer _____

**13** Round 125.0896 to the nearest thousandth.

WORKING

Answer _____

## Fractions

**14** Write $\frac{29}{6}$ as a mixed number.

WORKING

Answer _____

**15** Write $\frac{47}{8}$ as a mixed number.

WORKING

Answer _____

**16** Add: $\frac{3}{8} + \frac{7}{8}$

WORKING

Answer _____

**17** Add: $\frac{2}{3} + \frac{3}{4}$

WORKING

Answer _____

**18** Add: $3\frac{5}{16} + 2\frac{3}{4}$

WORKING

Answer _____

**19** Subtract: $\frac{7}{8} - \frac{5}{16}$

WORKING

Answer _____

**20** Subtract: $5\frac{1}{4} - 2\frac{11}{16}$

WORKING

Answer _____

**21** Multiply: $3\frac{3}{8} \times 6$

WORKING

Answer _____

**22** Divide: $\frac{5}{6} \div \frac{2}{9}$

WORKING

Answer _____

**23** Divide: $4\frac{1}{2} \div 6$

WORKING

Answer _____

## Decimals

**24** Add: 14.6 + 3.075 + 0.48

WORKING

Answer _____

**25** Add: 206.3 + 9.87 + 41.059

WORKING

Answer _____

**26** Subtract: 52 − 17.384

WORKING

Answer _____

**27** Subtract: 9.05 − 3.678

WORKING

Answer _____

**28** Multiply: 4.73 × 0.6

WORKING

Answer _____

**29** Multiply: 12.5 × 3.08

WORKING

Answer _____

**30** Divide: 17.3 ÷ 6. Round to the nearest thousandth.

WORKING

Answer _____

## Conversions

**31** Write $\frac{5}{8}$ as a decimal.

WORKING

Answer _____

**32** Write $\frac{11}{16}$ as a decimal rounded to the nearest thousandth.

WORKING

Answer _____

**33** Write 0.064 as a fraction in lowest terms.

WORKING

Answer _____

**34** Write 135% and 7.5% as decimals.

WORKING

Answer _____

**35** Write 0.068 and 2.4 as percents.

WORKING

Answer _____

**36** Write $1\frac{3}{5}$ and $\frac{7}{20}$ as percents.

WORKING

Answer _____

**37** Write 45% as a fraction in lowest terms.

WORKING

Answer _____

## Percent of a number

**38** Find 15% of 260.

WORKING

Answer _____

**39** Find 6% of 1,450.

WORKING

Answer _____

**40** Find 125% of 48.

WORKING

Answer _____